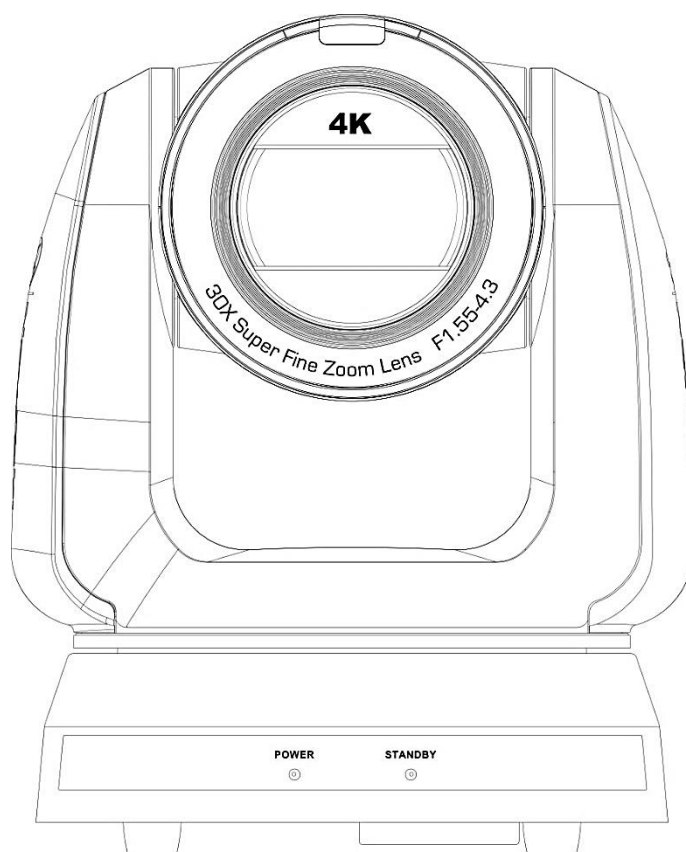


VC-A71P/VC-A71PN

4K PTZ Video Camera

User Manual - English



To download the latest version of Quick Start Guide, multilingual user manual, software, or driver, etc., please visit Lumens

<https://www.MyLumens.com/support>

Table of Contents

Copyright Information	2
Chapter 1 Safety Instructions	3
Chapter 2 Package Contents	4
Chapter 3 Function Introduction	5
3.1 I/O functions Introduction	5
3.2 Description of LED indicator	7
3.3 Tally Lamp Function Description	7
Chapter 4 Instruction for installation	8
4.1 Camera dimension	8
4.2 Preparation before installation	9
4.3 Instruction for installation	9
4.4 Connecting devices	14
Chapter 5 Remote Control and Setting Menu	17
5.1 Functions of remote control	17
5.2 Setting Menu	18
Chapter 6 Network Function Settings Description	29
6.1 Connecting Camera to Network	29
6.2 Web Page Function Description	33
Chapter 7 DIP Switch Setting	50
7.1 DIP SWITCH	50
Chapter 8 Troubleshooting	51
Supplier's Declaration of Conformity 47 CFR § 2.1077 Compliance Information	53

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Chapter 1 Safety Instructions

Always follow these safety instructions when using the product:

1 Operation

- 1.1 Please use the product in the recommended operating environment, away from water or source of heat.
- 1.2 Do not place the product in tilted position or unstable trolley, stand or table.
- 1.3 Please clean the dust on the power plug prior to usage. Do not insert the product's power plug into a multiplug to prevent sparks or a fire.
- 1.4 Do not block the slots and openings in the case of the product. They provide ventilation and prevent the product from overheating.
- 1.5 Do not open or remove covers, otherwise it may expose you to dangerous voltages and other hazards. Refer all servicing to licensed service personnel.
- 1.6 Unplug the product from the wall outlet and refer servicing to licensed service personnel when the following situations happen:
 - If the power cords are damaged or frayed.
 - If liquid is spilled into the product or the product has been exposed to rain or water.

2 Installation

- 2.1 For security considerations, please make sure the standard hanging rack you bought is in line with UL or CE safety approbations and installed by technician personnel approved by agents.

3 Storage

- 3.1 Do not place the product where the cord can be stepped on as this may result in fraying or damage to the lead or the plug.
- 3.2 Unplug the product during thunderstorms or if it is not going to be used for an extended period.
- 3.3 Do not place the product or accessories on top of vibrating equipment or heated objects.

4 Cleaning

- 4.1 Disconnect all the cables prior to cleaning and wipe the surface with a dry cloth. Do not use alcohol or volatile solvents for cleaning.

5 Batteries (for products or accessories with batteries)

- 5.1 When replacing batteries, please only use similar or the same type of batteries.
- 5.2 When disposing of batteries or products, please adhere to the relevant instructions in your country or region for disposing of batteries or products.

■ Precautions

	This symbol indicates that this equipment may contain dangerous voltage which could cause electric shock. Do not remove the cover (or back). No user-serviceable parts inside. Refer servicing to licensed service personnel.		This symbol indicates that there are important operating and maintenance instructions in this User Manual with this unit.
---	---	--	---

■ FCC Warning

This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment.

Notice:

The changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

■ IC Warning

This digital apparatus does not exceed the Class A limits for radio noise emissions from digital apparatus as set out in the interference-causing equipment standard entitled "Digital Apparatus," ICES-003 of Industry Canada.

Cet appareil numérique respecte les limites de bruits radioélectriques applicables aux appareils numériques de Classe A prescrites dans la norme sur le matériel brouilleur: "Appareils Numériques," NMB-003 édictée par l'Industrie.

■ EN55032 CE Warning

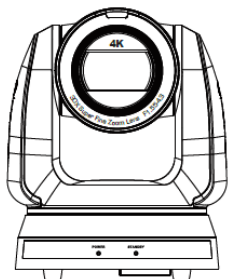
Operation of this equipment in a residential environment could cause radio interference.

■ KC Warning

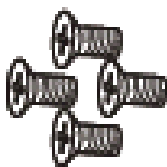
This equipment is Industrial (Class A) electromagnetic wave suitability equipment and seller or user should take notice of it, and this equipment is to be used in the places except for home.

Chapter 2 Package Contents

VC-A71P/VC-A71PN



M3 Screws

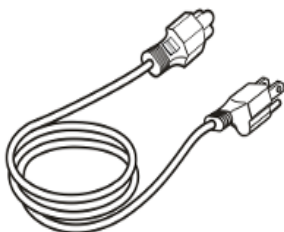


Silver x8 / Black x2

Remote Control

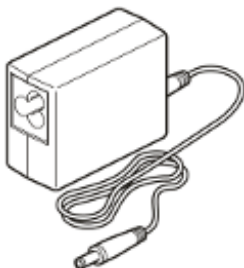


Power Cord

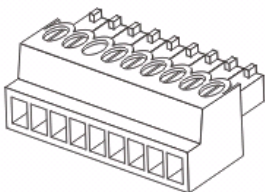


Appearance may vary
depending on
country/region

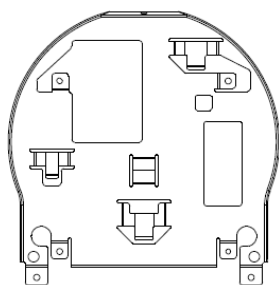
Power Adapter



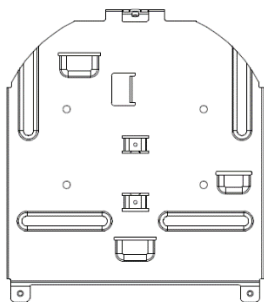
RS-422 Connector



Metal Plate A



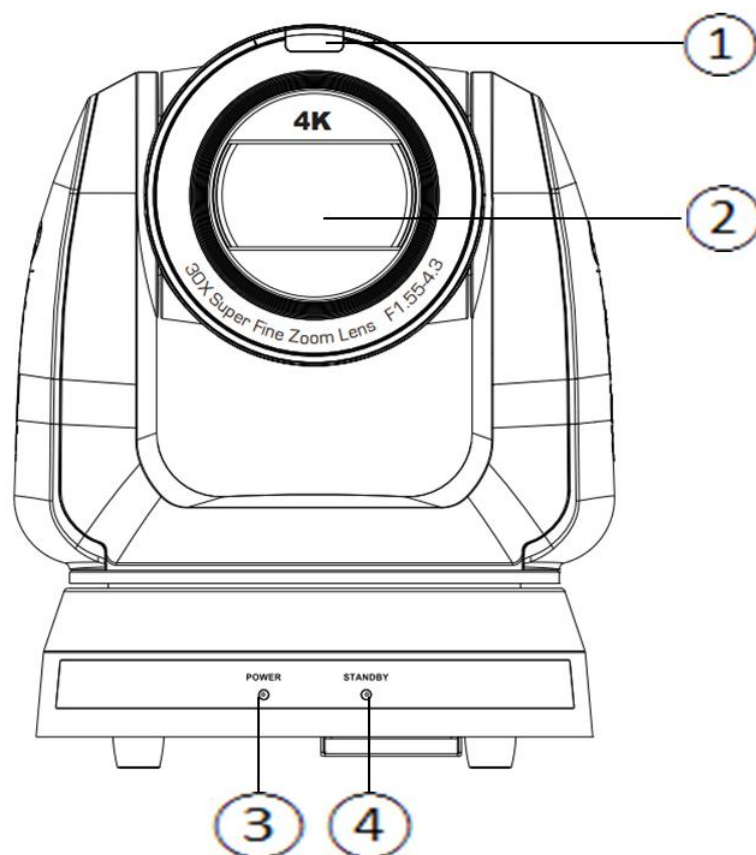
Metal Plate B



Chapter 3 Function Introduction

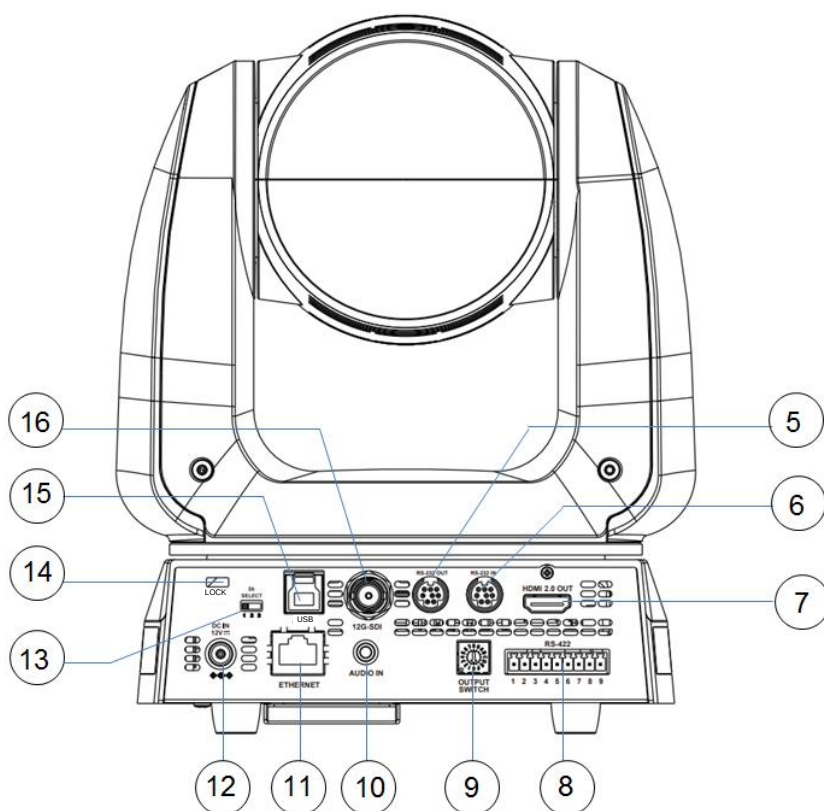
3.1 I/O functions Introduction

3.1.1 Front View



NO.	Item	Function Descriptions
1.	Tally indicator light	Displays the tally light status of the camera
2.	Camera lens	30x HD camera lens
3.	Power LED indicator	Displays the status of the camera. Please refer to 3.2 Description of LED indicator
4.	Standby LED indicator	Displays the status of the camera. Please refer to 3.2 Description of LED indicator

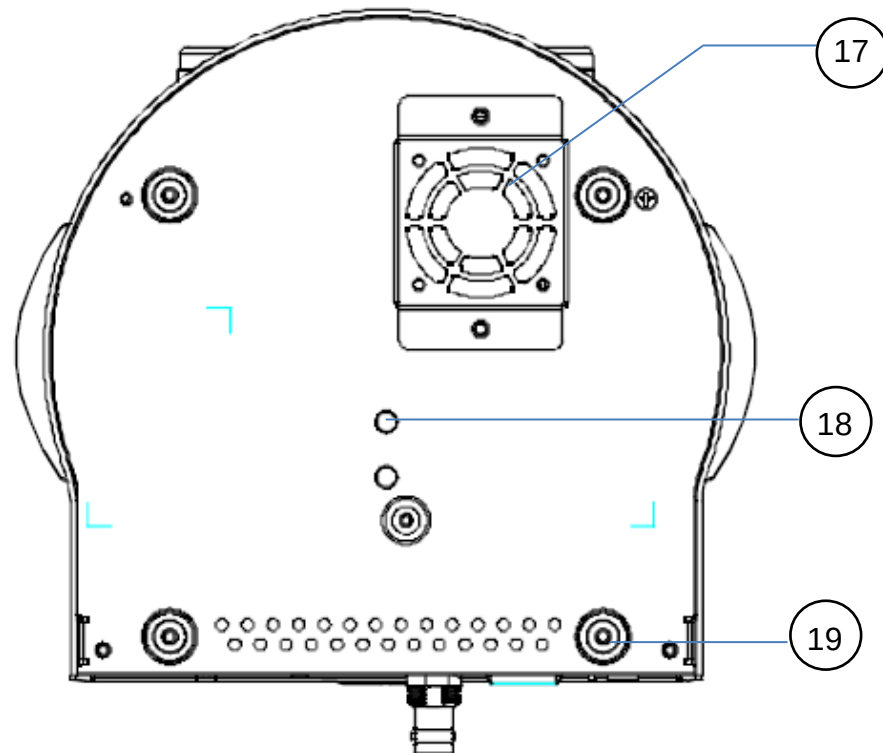
3.1.2 Back View



<Remark> The figure is an example of VC-A71P. VC-A71PN does not support SDI and USB output.

NO.	Item	Function Descriptions
5.	RS-232 output	Up to 7 cameras can be connected in a serial connection
6.	RS-232 input	Up to 7 cameras can be connected in a serial connection
7.	HDMI 2.0 output	HDMI 2.0 output (Audio output supported)
8.	RS-422 port	Up to 7 cameras can be connected in a serial connection
9.	OUTPUT SWITCH	Adjust the resolution setting. The default is 3840 x 2160/59.94p
10.	Audio input	Support Line In/Mic In
11.	Network port	Support routers or hubs with power supply ■ VC-A71P : PoE++ (IEEE802.3bt) ■ VC-A71PN : PoE+ (IEEE802.3at)
12.	DC 12 V power connector	AC power supply connecting port
13.	IR SELECT	The camera can be set from IR1 to IR3. To control the camera, the selected IR ID must match the Camera Select setting on the remote control.
14.	Kensington lock hole	Safety lock slot
15.	USB 3.0 port	Used to connect to a computer for USB video output. <Remark> Support up to 3840 x 2160/29.97p <Remark> VC-A71PN does not support this output port.
16.	12G-SDI output	12G-SDI output (Audio output supported) <Remark> VC-A71PN does not support this output port.

3.1.3 Bottom



NO.	Item	Function Descriptions
17.	Mute cooling fan	The cooling fan automatically adjusts its rotation speed based on the operating temperature of the camera. <Remark>The cooling fan is not available on VC-A71PN.
18.	Tripod deck lock hole	Compatible with standard 1/4"-20 UNC tripod mounts.
19.	Rotatable foot pad	When installing the accessory metal plates, remove the four rotatable foot pads on the bottom by turning them counterclockwise, then install the metal plates.

3.2 Description of LED indicator

Status	Power	Standby
Startup in progress (initialization)	Green light	Orange light
In use	Green light	No indicator
In standby mode	No indicator	Orange light

3.3 Tally Lamp Function Description

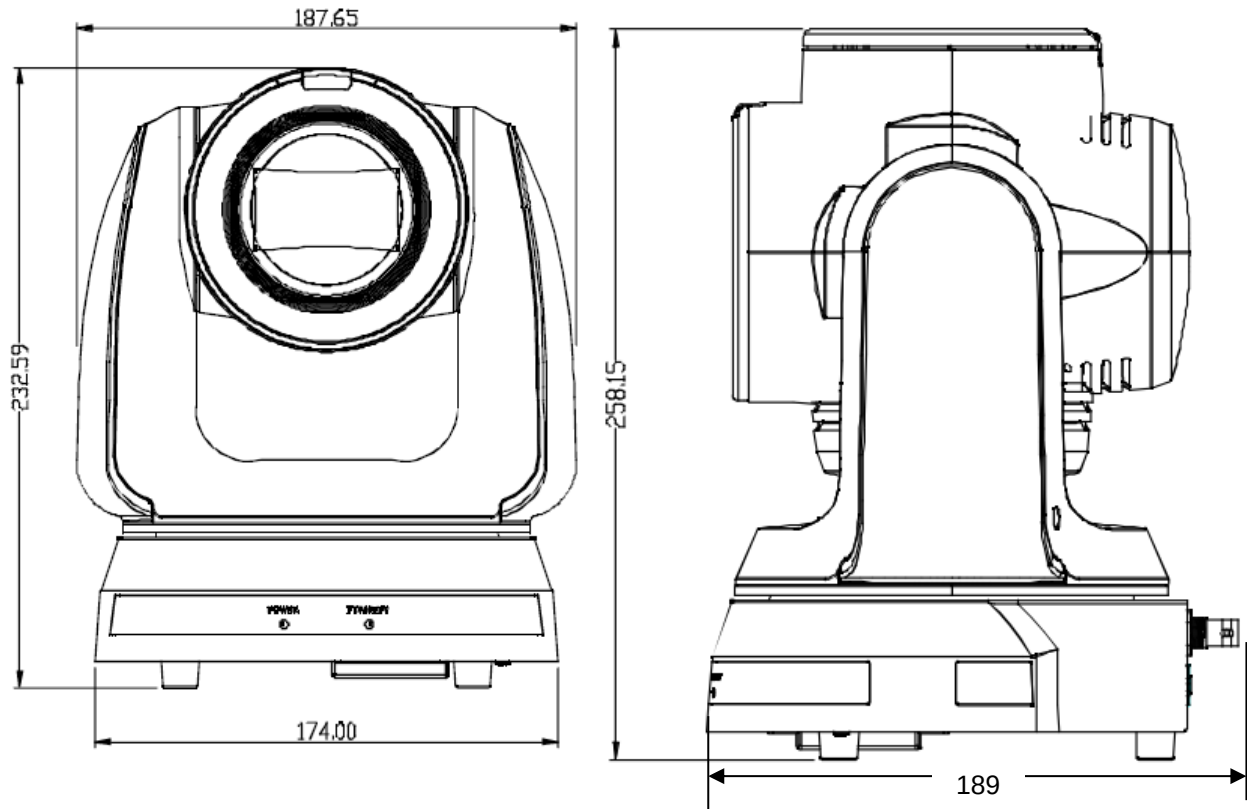
3.3.1 The Tally Lamp function can be enabled via RS-232 command. Please refer to the RS-232 command set document for details.

Chapter 4 Instruction for installation

4.1 Camera dimension

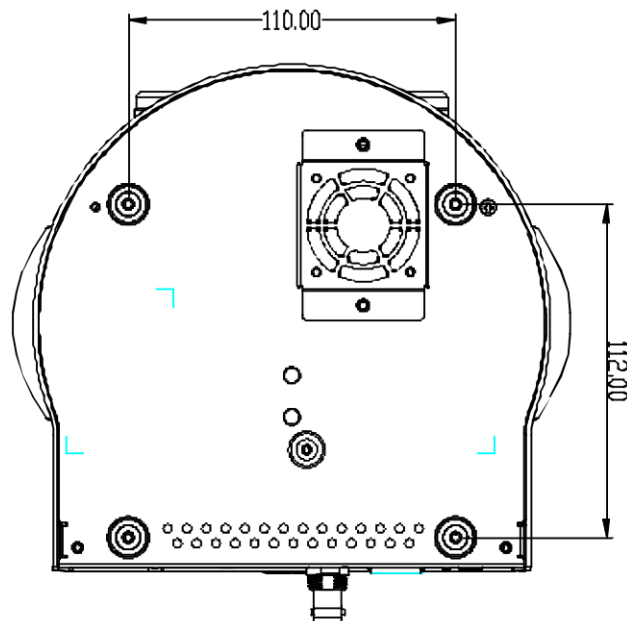
4.1.1.1 Camera front view and side view

- Length x width x height: 232 x 188 x 189 mm
- Weight: 3.0Kg (excluding the metal plate)



4.1.1.2 Camera base

The camera base includes tripod mounting holes and is compatible with standard 1/4"-20 UNC tripod mounts.



4.2 Preparation before installation

Installation and connection of the HD camera requires special skills. To install by yourself, please follow necessary steps, ensure steady and tight installation of the device, and pay attention to your safety to avoid any accident.

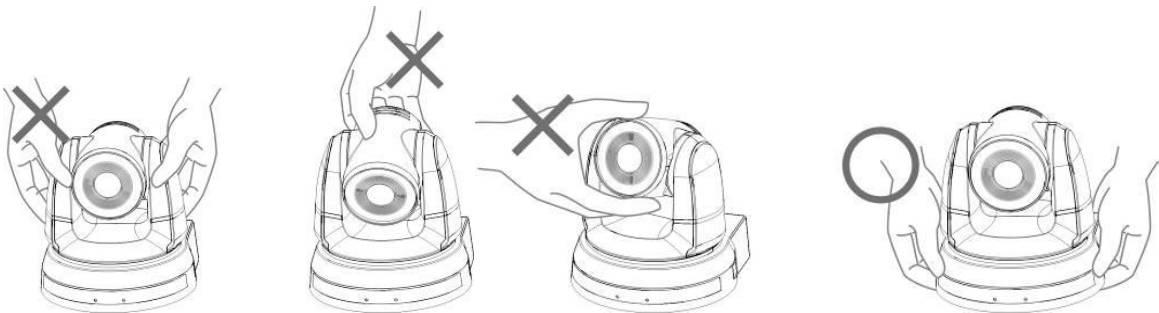
- 4.2.1 Ensure the safety of the installation environment. Please do not install the device on a unstable ceiling or in a place where the device is in danger of falling to avoid any accident.
- 4.2.2 Please check whether accessories in the box are complete or not. Please contact the supplier for any shortage, and make sure to keep the accessories in the box intact.
- 4.2.3 Please choose a proper place for installation of camera in advance. Please determine an installation place according to the following requirements
 - 4.1.1.1 Confirm the position for the object to be captured.
 - 4.1.1.2 Confirm whether the camera is set at a proper distance from other light sources.

4.3 Instruction for installation

4.3.1 I would like to install camera on the desk

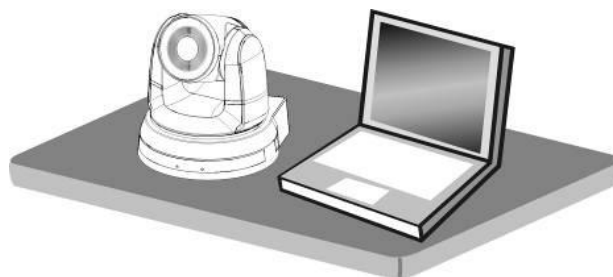
4.3.1.1 Precautions for installation

- ◆ Please install the machine on a flat desk
- ◆ Do not grab the camera head by hand when handling the device
- ◆ Do not rotate the camera head by hand. Improper rotation may result in breakdown of the camera



4.3.1.2 Installation steps

1. Place the camera on a flat desk directly to ensure the normal vertical and horizontal operation of the machine



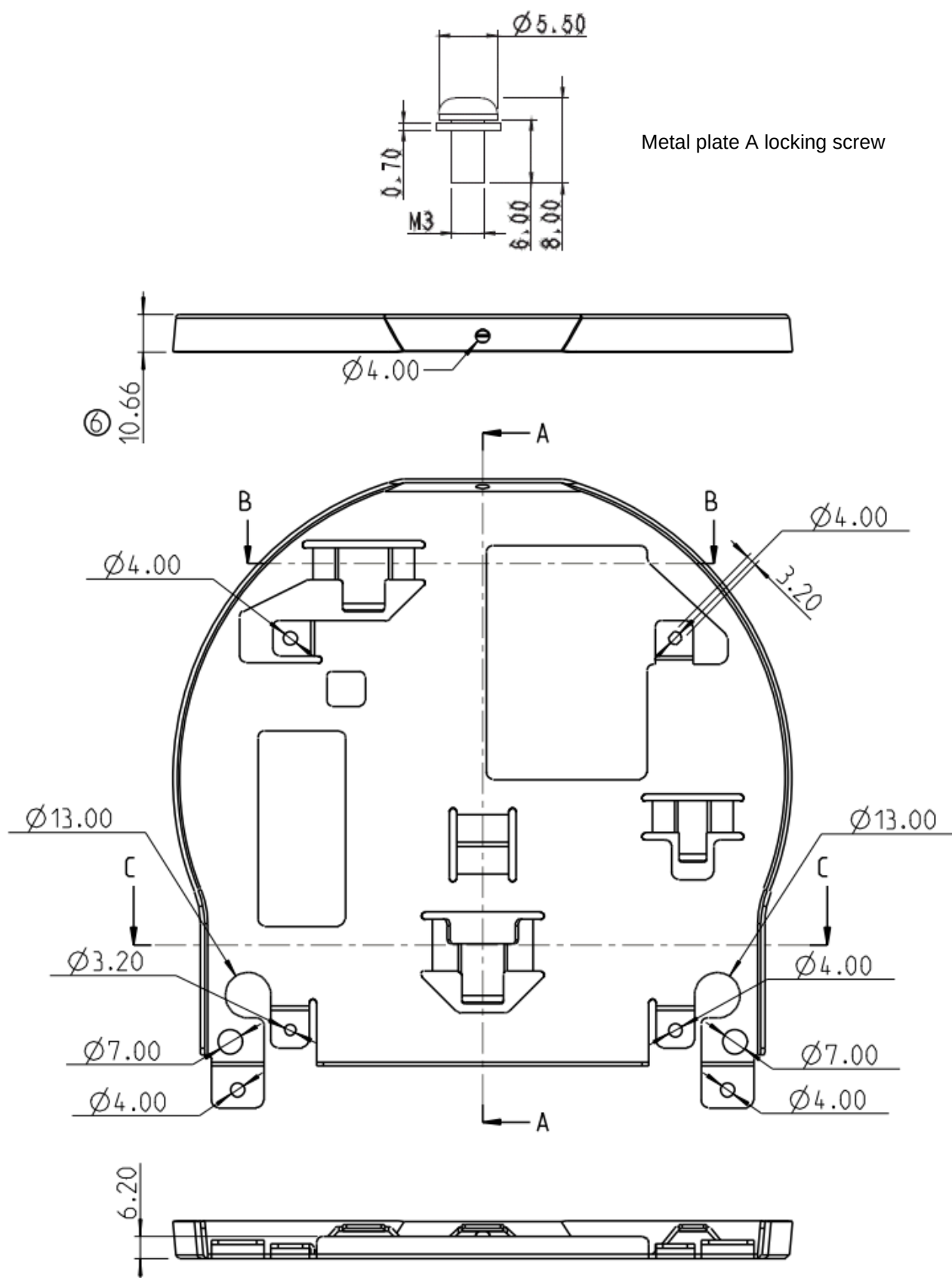
4.3.2 I would like to install the camera on the ceiling

4.3.2.1 Prepare for the parts and equipment required during the installation

1. Accessories in the box (metal plates A, B, M3 screw silver x 8, black x 2)
2. Screw for locking on ceiling mounted hanger x 4
3. Drilling machine, screw driver, ladder

4.3.2.2 Metal Plate size diagram

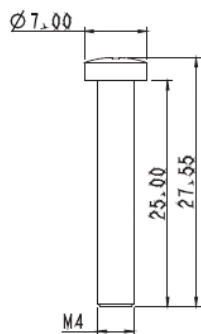
■ Metal plate A - machine side



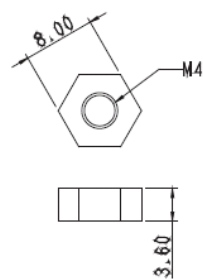
Metal plate A locking screw

Metal plate A - machine side

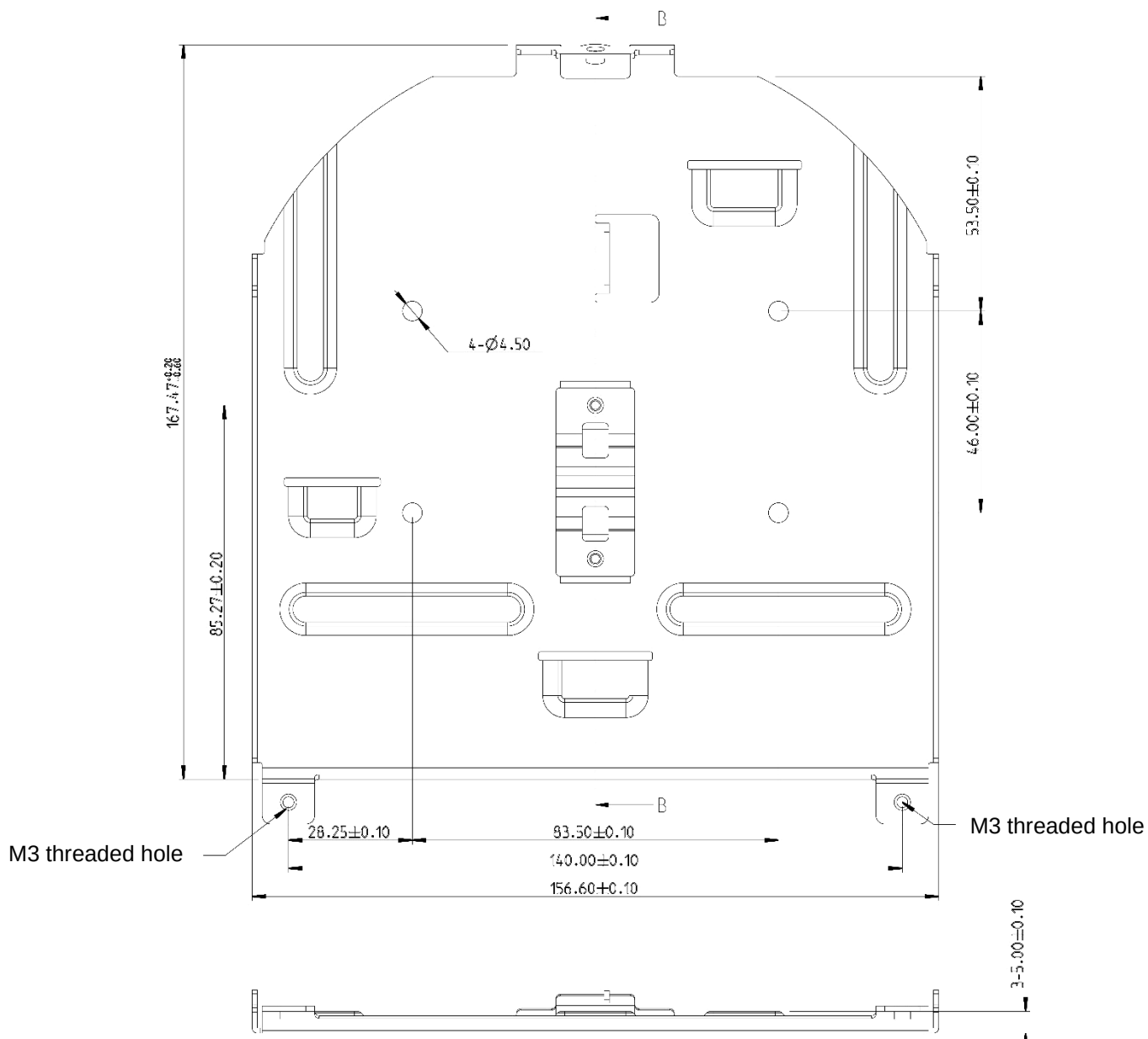
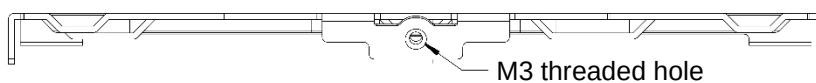
■ Metal plate B - ceiling side



Metal plate B locking screw



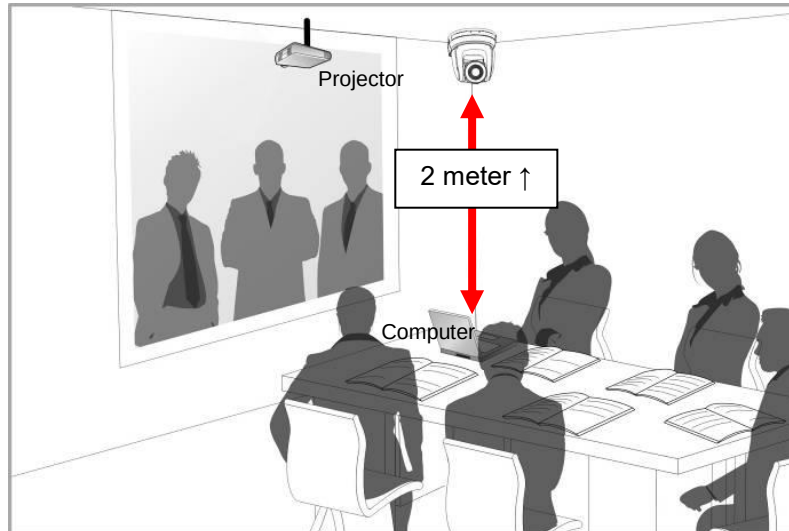
Metal plate B locking bolt



Metal plate B - ceiling side

4.3.2.3 Precautions for installation

1. Before installation, please confirm the orientation of the machine relative to the object to be captured
2. It is recommended that the machine should be set at a distance of more than 2 meters away from the object to be captured. Please adjust for a best distance according to the magnification of the lens

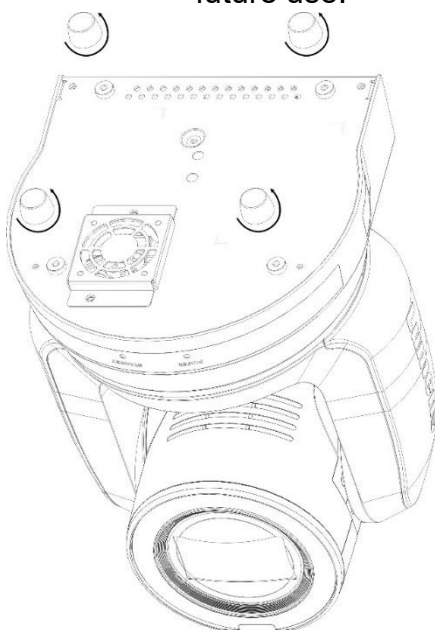


3. The machine (including metal plates) is weighed at about **3** kg. If it is to be installed on the ceiling, please use the hanger that has obtained UL security approval to prevent the machine from falling down.
4. Please check whether the camera is installed securely on a regular basis.
5. Please clean the cooling fan located at the bottom to ensure the good heat dissipation effect of the machine on a regular basis.

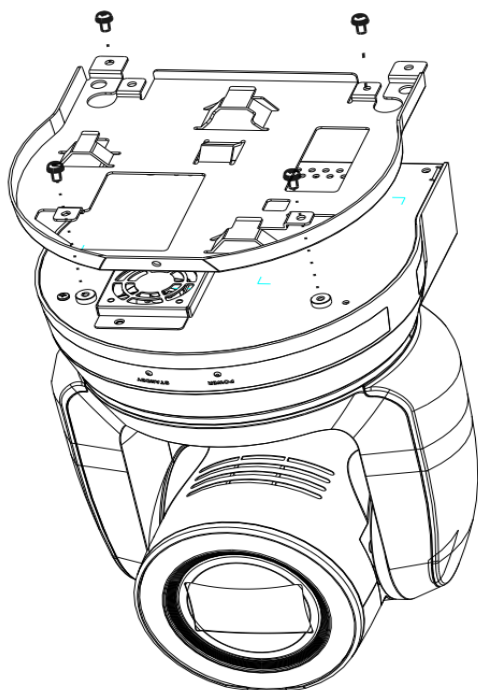
4.3.2.4 Installation steps

1. Remove 4 pieces of foot pads on the bottom of the camera by rotating the foot pads counterclockwise.

<Remark> Please keep the foot pads properly after removal and store them for future use.



2. Fix the metal plate A on the machine base with 4 M3 silver screws



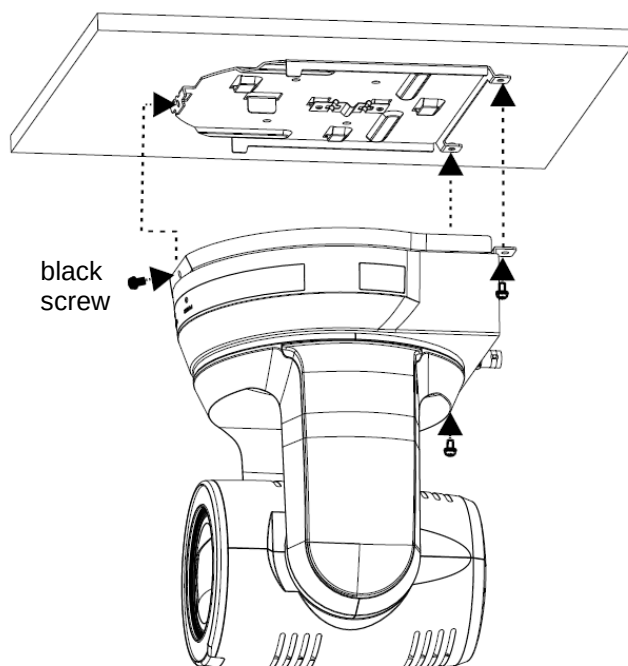
3. Lock the metal plate B on ceiling mounted hanger

<Caution>:

- (1) Please use the hanger that has obtained UL security approval
- (2) Please reserve the hole for the connecting wires of the camera

4. Combine the metal plate A and the metal plate B

- (1) Push the metal plate A up to the ceiling and then to the right to latch the metal plate B
- (2) And then secure with 2 M3 silver screws and 1 M3 black screw



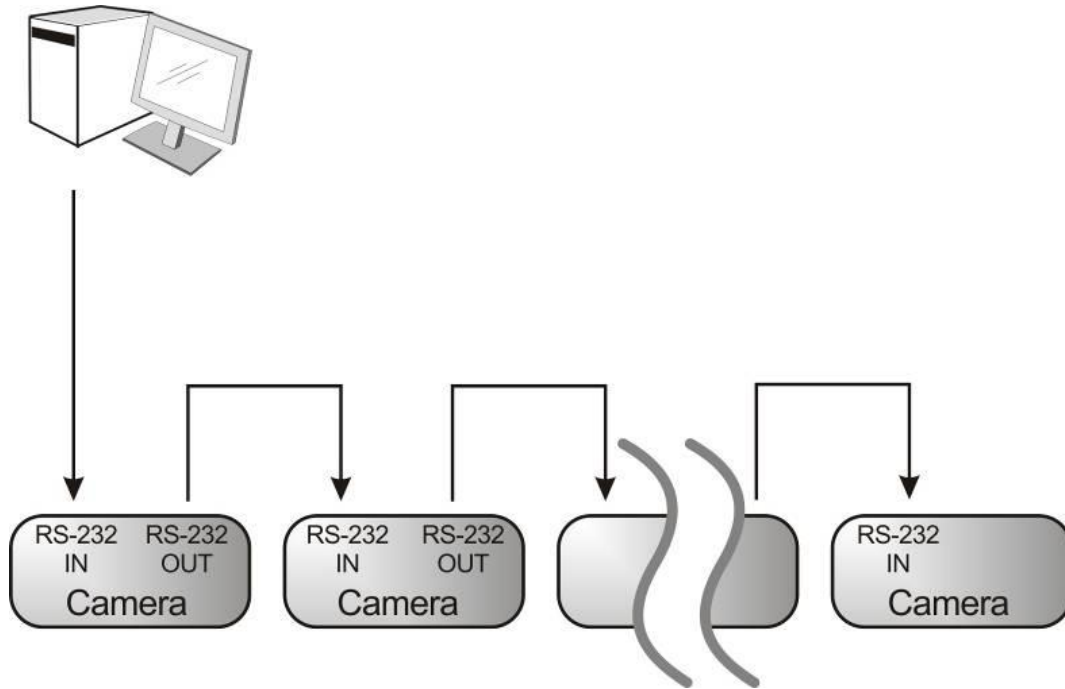
4.3.2.5 How to remove

1. Remove the connecting wires from the camera
2. Uninstall the camera together with the ceiling, loosen the three screws that fix the metal plates A and B and push to the left to remove the machine
3. Then remove the screws on the hanger and the machine

4.4 Connecting devices

4.4.1 Connecting RS-232

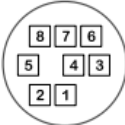
With RS-232 in/out, at most 7 Lumens cameras can be connected.



■ RS-232 pins definition instructions

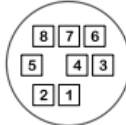
▼ RS-232 IN Pins Instructions

NO	Pins	Signals
1	DTR	Data Transmission Reade
2	DSR	Data Set Reade
3	TXD	Transmit Data
4	GND	Ground
5	RXD	Receive Data
6	GND	Ground
7	IR OUT	IR Commander Signal
8	N.C.	No Connection



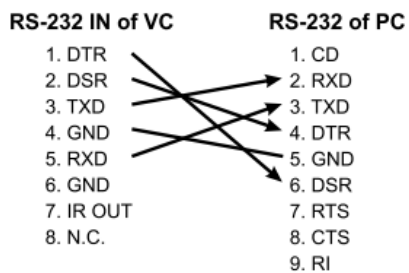
▼ RS-232 OUT Pins Instructions

NO	Pins	Signals
1	DTR	Data Transmission Reade
2	DSR	Data Set Reade
3	TXD	Transmit Data
4	GND	Ground
5	RXD	Receive Data
6	GND	Ground
7	OPEN	Reserved
8	OPEN	Reserved



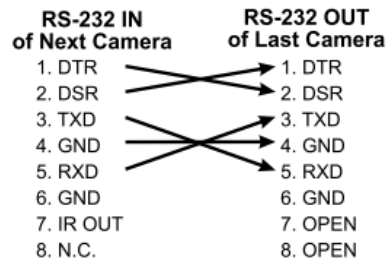
▼ RS-232 Connection Instructions

RS-232 IN of VC	RS-232 of PC
1. DTR	1. CD
2. DSR	2. RXD
3. TXD	3. TXD
4. GND	4. DTR
5. RXD	5. GND
6. GND	6. DSR
7. IR OUT	7. RTS
8. N.C.	8. CTS
	9. RI



▼ RS-232 Connection Instructions

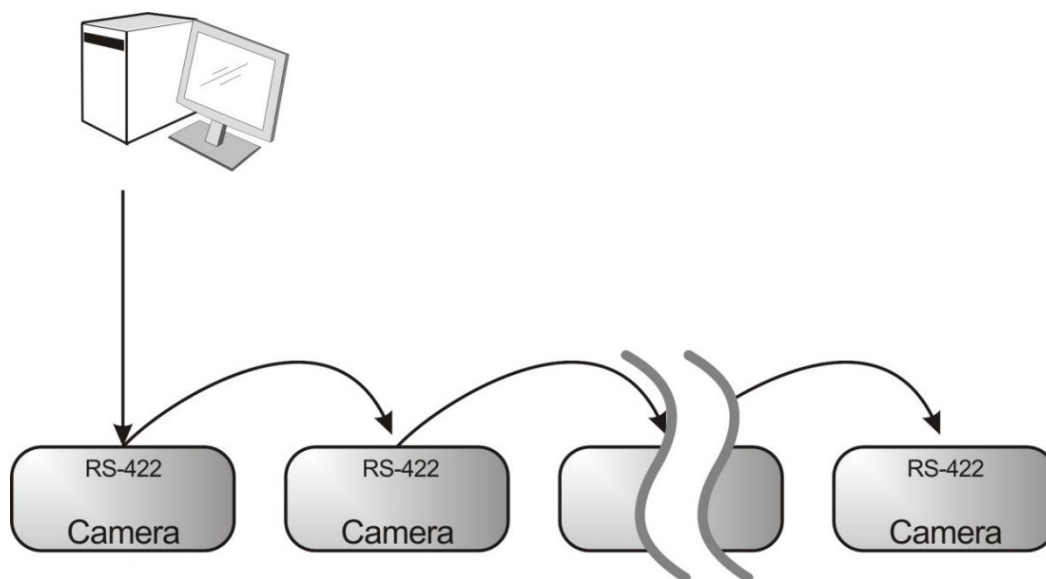
RS-232 IN of Next Camera	RS-232 OUT of Last Camera
1. DTR	1. DTR
2. DSR	2. DSR
3. TXD	3. TXD
4. GND	4. GND
5. RXD	5. RXD
6. GND	6. GND
7. IR OUT	7. OPEN
8. N.C.	8. OPEN



4.4.2 Connecting RS-422

With RS-422, up to 7 Lumens cameras can be connected.

<Caution> When RS-422 connection is being used, do not use RS-232 connection.



■ RS-422 pins definition instructions

▼ RS-422 Pins Instructions

RS-422

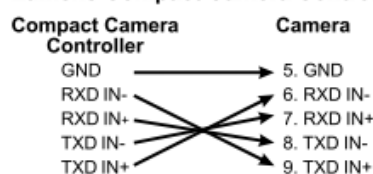
Pin NO.	Function
1	RXD OUT-
2	RXD OUT+
3	TXD OUT-
4	TXD OUT+
5	GND
6	RXD IN-
7	RXD IN+
8	TXD IN-
9	TXD IN+

▼ RS-422 Serial Connection

Up to seven cameras can be daisy-chained

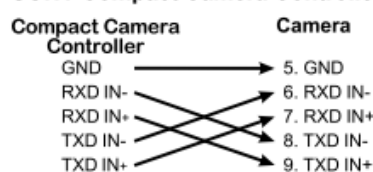
▼ RS-422 Connection Instructions

—Applicable to
Lumens Compact Camera Controller



▼ RS-422 Connection Instructions

—Applicable to
SONY Compact Camera Controller



Compact Camera Controller

1	NC
2	NC
3	NC
4	NC
5	GND
6	RXD IN-
7	RXD IN+
8	TXD IN-
9	TXD IN+

Third Camera

1	RXD OUT-
2	RXD OUT+
3	TXD OUT-
4	TXD OUT+
5	GND
6	RXD IN-
7	RXD IN+
8	TXD IN-
9	TXD IN+

Second Camera

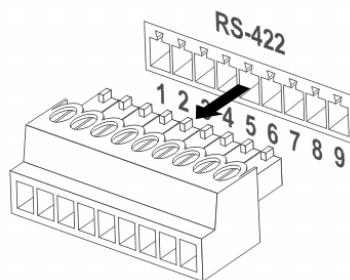
1	RXD OUT-
2	RXD OUT+
3	TXD OUT-
4	TXD OUT+
5	GND
6	RXD IN-
7	RXD IN+
8	TXD IN-
9	TXD IN+

First Camera

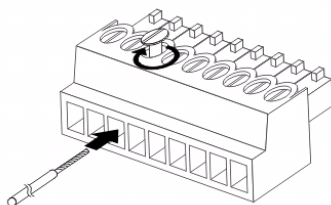
1	RXD OUT-
2	RXD OUT+
3	TXD OUT-
4	TXD OUT+
5	GND
6	RXD IN-
7	RXD IN+
8	TXD IN-
9	TXD IN+

■ RS-422 connection instructions

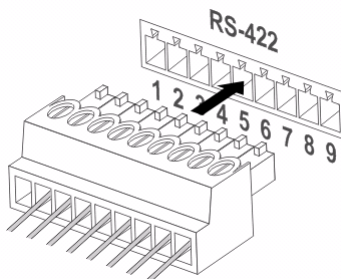
1. Hold the two sides of RS-422 connector and pull out in the direction shown by the arrow in the figure below



2. Peel off a section of copper wire (AWG Nos. 28 to 18) and insert it into the connector hole; then use flat screw driver to fix it

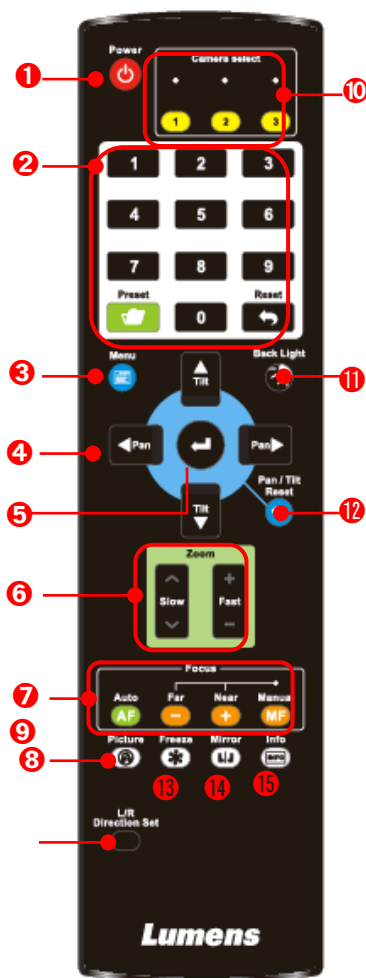


3. Insert the wired RS-422 connector back to the Camera. Now the connection is completed



Chapter 5 Remote Control and Setting Menu

5.1 Functions of remote control



No.	Item	Description
1.	Power	Power Switch
2.	Preset	■ Numeric Keys 0 - 9: Press a number button to move the camera to the corresponding saved position. ■ Preset Position Assignment: Assign a preset number to a desired position. (Hold the <i>Preset</i> button, then press the desired number to save.) ■ Preset Position Deletion Delete a saved position. (Hold the <i>Reset</i> button then press the desired number to delete.)
3.	Menu	Display OSD menu
4.	Pan/ Tilt	Move the lens
5.	Enter	Confirm a selection / Move the camera to the home position
6.	Zoom	■ Fast: Zooming in / out fast ■ Slow: Zooming in / out slowly
7.	Focus	■ Auto: Auto Focus ■ Manual: Enable Manual Focus ■ Far/ Near: Adjust the focal length in manually
8.	Picture	Switch image effect (OFF / Neg / B&W)
9.	L/R Direction Set	Reverse the pan and tilt direction
10.	Camera select	Choose Camera ID from 1 ~ 3
11.	Back Light	Turn on/ off back light compensation
12.	Pan/ Tilt Reset	Calibrate the Pan and Tilt data.
13.	Freeze	Freeze the screen
14.	Mirror	Rotate the image (OFF / Mirror / Flip / Mirror+Flip)
15.	Info	Status information

5.3 Setting Menu

<Remark>Press [Menu] on the **remote control** to enter the setting menu; the **bold underlined** values in the following table are defaults.

1st Level Major Items	2nd Level Minor Items	3rd Level Adjustment Values		Function Descriptions
Exposure	Mode	1. <u>Full Auto</u> 2. Shutter Pri 3. Iris Pri 4. Manual		Exposure mode setting
	Exposure Comp.	On / <u>Off</u>		AE Level
	Exposure Comp. Level	-5 ~ <u>C</u> ~ 5		The value can be adjusted after Exposure Comp. is activated
	Spot Light	On / <u>Off</u>		
	Spot Light Position	X (0 ~ 6) Y (0 ~ 4)		The value can be adjusted after Spot Light is activated
	Shutter Pri	60/30 mode	50/25 mode	Adjustable when the Exposure mode is set to Shutter Pri
		1/10000	1/10000	
		1/5000	1/5000	
		1/3000	1/3000	
		1/2500	1/2500	
		1/2000	1/1750	
		1/1500	1/1250	
		1/1000	1/1000	
		1/725	1/600	
		1/500	1/425	
		1/350	1/300	
		1/250	1/215	
		1/180	1/150	
		1/120	1/120	
		1/100	1/100	
		1/90	1/75	
		<u>1/60</u>	<u>1/50</u>	
		1/30	1/25	
		1/15	1/12	
		1/8	1/6	
		1/4	1/3	
		1/2	1/2	
		1/1	1/1	

1st Level Major Items	2nd Level Minor Items	3rd Level Adjustment Values		Function Descriptions
		1/120	1/120	
		1/100	1/100	
		1/90	1/75	
		<u>1/60</u>	<u>1/50</u>	
		1/30	1/25	
		1/15	1/12	
		1/8	1/6	
		1/4	1/3	
		1/2	1/2	
		1/1	1/1	
	Manual Iris	1. F1.6 2. F2 3. <u>F2.2</u> 4. F2.7 5. F3.2 6. F3.8 7. F4.5 8. F5.4 9. F6.3 10.F7.8 11. F9 12. F11 13. F13 14. F16 15. F18 16. Off		Manually set the iris
		Gain Limit	1. 9 dB 2. 12 dB 3. 15 dB 4. 18 dB 5. 21 dB 6. 24 dB 7. 27 dB 8. 30 dB 9. 33 dB 10. 36 dB 11. 39 dB 12. 42 dB 13. 45 dB	

1st Level Major Items	2nd Level Minor Items	3rd Level Adjustment Values	Function Descriptions
	Iris Limit	1. <u>F2.2</u> 2. F2.7 3. F3.2 4. F3.8 5. F4.5 6. F5.4 7. F6.3 8. F7.8 9. F9 10. F11	Max. limit value of iris
	WDR	1. <u>Off</u> 2. 1 3. 2 4. 3	
	Smart AE	5. On / <u>Off</u>	Available when "Smart AF" is on
White Balance	Mode	1. <u>Auto</u> 2. Indoor 3. Outdoor 4. One Push WB 5. ATW 6. Manual 7. Sodium Lamp 8. 3000K 9. 4300K 10. 5000K 11. 6500K 12. 8000K	Select the color temperature mode 1. 4000k ~ 7000k 2. 3200k 3. 5800k 4. 1700k ~ 10000k 5. 1700k ~ 10000k 6. Custom 7. 2800k
	One Push Trigger	<u>ENTER</u>	One push trigger
	Manual Red	0 ~ <u>C</u> ~ 128	Adjustable when the white balance mode is set to Manual
	Manual Blue	0 ~ <u>C</u> ~ 128	Adjustable when the white balance mode is set to Manual
Picture	Picture Effect	1. <u>Off</u> 2. Neg 3. B & W	
	Sharpness	0 ~ <u>A</u> ~ 14	

1st Level Major Items	2nd Level Minor Items	3rd Level Adjustment Values	Function Descriptions
	2D NR	1. Off 2. <u>1</u> 3. 2 4. 3	
	3D NR	1. Off 2. LOW 3. <u>Typ</u> 4. Max	
	Image Mode	1. <u>Standard</u> 2. LED Mode 3. Custom	The user may customize the desired image mode.
	Image Mode Load	<u>ENTER</u>	Adjustable when the Image mode is set to Custom . After selected, the corresponding Image mode parameters will be read and applied to Custom
	Brightness	0 ~ <u>A</u> ~ 15	Adjustable when the Image mode is set to Custom
	Hue	0 ~ <u>A</u> ~ 15	Adjustable when the Image mode is set to Custom
	Saturation	0 ~ <u>A</u> ~ 15	Adjustable when the Image mode is set to Custom
	Gamma	0 ~ <u>3</u>	Adjustable when the Image mode is set to Custom
	Black Level	0~ <u>Auto</u> ~10	Adjust the brightness of dark areas.
Pan Tilt Zoom	Pan/Tilt Limit	On / <u>Off</u>	Enable or Disable the angle limit
	Pan Right Limit	0 ~ <u>170</u>	Set the maximum right pan angle
	Pan Left Limit	<u>-170</u> ~ 0	Set the maximum left pan angle
	Tilt UP Limit	0 ~ <u>90</u>	Set the maximum upward tilt angle
	Tilt Down Limit	<u>-30</u> ~ 0	Set the maximum downward tilt angle
	Pan Flip	On / <u>Off</u>	Reverse the Pan direction

1st Level Major Items	2nd Level Minor Items	3rd Level Adjustment Values	Function Descriptions
	Tilt Flip	On / <u>Off</u>	Reverse the Tilt direction
	Preset Speed	1. 5 deg/sec 2. 25 deg/sec 3. 50 deg/sec 4. 80 deg/sec 5. 120 deg/sec 6. <u>160 deg/sec</u> 7. 200 deg/sec 8. 300 deg/sec	Set the rotation speed of the cradle head when a Preset position is triggered.
	PTZ Speed Comp	On / <u>Off</u>	Automatically adjust pan/tilt speed based on the zoom level
	Preset Mode	<u>Default</u> / PTZ Setting	▪ Default: After a preset is recalled, all saved parameters including Pan/Tilt/Zoom/Focus/picture will all be applied. ▪ PTZ Setting: After a preset is recalled, only the Pan/Tilt/Zoom/Focus parameter will be applied.
	D-Zoom Limit	<u>x1</u> ~x12	Set the D-zoom limit <Remark>3840 x 2160 59.94/50 does NOT support this setting
	PTZ Motion Sync	On / <u>Off</u>	Off: The camera first moves to the preset position, then zooms in or out. On: The camera pans or tilts while zooming in or out simultaneously, allowing it to reach the final position faster. <Remark> This function uses the default speed setting, but the actual speed may be automatically adjusted based on the PTZ movement range.

1st Level Major Items	2nd Level Minor Items	3rd Level Adjustment Values	Function Descriptions
D-Effect	Mirror / Flip	1. <u>Off</u> 2. MIRROR 3. Flip 4. Mirror + Flip	Select how the image is mirrored or flipped.
Auto Focus	AF Sensitivity	1. LOW 2. <u>Middle</u> 3. High	The more sensitive the AF is, the faster it is triggered
	AF Frame	1. Center Area 2. Full Area 3. <u>Auto</u>	Center Area: The camera focuses only on the center area. Full Area: The camera focuses on the entire image. Auto: The camera automatically selects the focus area.
	Zoom Tracking	Manual/ <u>Curve Tracking</u> / Zoom Tracking	This function can only be enabled in the Manual Focus (MF) mode Set the focus mode during lens zooming in Manual Focus mode
	PTZ Assist	On / <u>Off</u>	This function can be enabled only in Manual Focus (MF) mode. After the PTZ Assist is enabled, the camera perform focus once during PTZ moving. Changes in the scene afterward will not trigger the focusing.
	SMART AF	On / <u>Off</u>	Enable this function and AF will primarily focus on the human face.
Ethernet	DHCP	<u>On</u> / Off	Enable/Disable DHCP setting using left and right arrow keys and press [ENTER] to apply setting.

1st Level Major Items	2nd Level Minor Items	3rd Level Adjustment Values	Function Descriptions
	IP Address	<u>192.168.100.100</u>	Press [ENTER] to enter edit mode. Use the Up and Down keys to select an item, then adjust the value using the Left and Right keys or the numeric keys.
	Subnet Mask	<u>255.255.255.0</u>	Press [ENTER] to enter edit mode. Use the Up and Down keys to select an item, then adjust the value using the Left and Right keys or the numeric keys.
	Gateway	<u>192.168.100.254</u>	Press [ENTER] to enter edit mode. Use the Up and Down keys to select an item, then adjust the value using the Left and Right keys or the numeric keys.
Audio	Audio In	<u>Line In</u> / Mic In	Select audio source type
	Audio Enable	On / <u>Off</u>	Turn on/off audio output
	Audio Volume	0 ~ <u>A</u> ~ 10	Volume Setting
	Audio Delay	On / <u>Off</u>	When audio and video are out of sync, enable this feature to set the audio delay time <Remark>For VC-A71PN, to enable this function, please turn off NDI HX
	Audio Delay Time(ms)	<u>-1</u> ~-500ms	Set audio delay time <Remark> For VC-A71PN, to enable this function, please turn off NDI HX
	Encode Type	1. <u>AAC</u> 2. G.711	Set encode type <Remark> For VC-A71PN, to enable this function, please turn off NDI HX

1st Level Major Items	2nd Level Minor Items	3rd Level Adjustment Values	Function Descriptions
	Encode Sample Rate	1. <u>48 KHz (AAC)</u> 2. 44.1 KHz (AAC) 3. 16 KHz (AAC) 4. 16 KHz (G.711) 5. 8 KHz (G.711)	Set the encode type and sample rate <Remark> SDI supports the audio output of 48 KHz only <Remark> For VC-A71PN, to support this function, please turn off NDI HX
System	Prompt	On / <u>Off</u>	Turn on/off the prompt information on the display
	IR Receive	<u>On</u> / Off	When it is disabled, the remote controller will not be able to control the camera. Reboot the camera can restore the function.
	Tally Lamp	On / <u>Off</u>	Enable or disable the Tally Lamp.

1st Level Major Items	2nd Level Minor Items	3rd Level Adjustment Values	Function Descriptions
	Tally CMMD Mode	Normal/ <u>Link</u>	Set whether the indicator light is linked with RS232 command ■ General (Not linked): The indicator light switch and mode shall be set separately. ➤ When the indicator light is on and the default mode is off, additional command is required to set the indicator light mode ➤ Setting indicator light mode when the indicator light is off will not trigger the indicator light ■ Link: ➤ When the indicator light is on, the default mode is red light Setting indicator light mode when the indicator light is off will turn on the indicator light
	Language	<u>English</u> / Chinese	Select a language.
	Initial Position	<u>Last Memory</u> / 1st Preset	Decide whether the camera will return to last saved position or the first preset position. <Remark> Default Initial position = Preset 0
	Control Device	Encoder / <u>Controller</u>	Control Device setting, Controller: for joystick Encoder: for tracking system

1st Level Major Items	2nd Level Minor Items	3rd Level Adjustment Values	Function Descriptions
	Motionless Preset	On / <u>Off</u>	When the function is enabled, the screen will Freeze when Preset is executed. Freeze will be released after Preset is completed.
	Control Port	<u>RS-232</u> / RS-422	Choose whether the control interface used is RS-232 or RS-422
	Protocol	<u>VISCA</u> / Pelco D	Select the protocol
	Baud Rate	<u>9600</u> / 38400	Choose the transmission speed of the control signal
	VISCA Address	<u>0</u> ~ 7	Assigned a camera ID to the camera. Select 0 for automated assignment.
	PELCO D Address	<u>1</u> ~ 255	Assigned a camera ID to the camera.
	Output Mode	1. <u>3840 x 2160/59.94p</u> 2. 3840 x 2160/50p 3. 3840 x 2160/29.97p 4. 3840 x 2160/25p 5. 1080p/59.94 6. 1080p/50 7. 1080p/29.97 8. 1080p/25 9. 720p/59.94 10. 720p/50 11. 720p/29.97 12. 720p/25 13. 1080i/59.94 14. 1080i/50	Choose the output resolution <Remark> VC-A71PN does not support 720p/29.97 and 720p/25 output modes <Remark> For HDMI/SDI output, when 1080i mode is set, NDI output is 1080p
	Color Space	Standard Profile/ rec.709/ BT.2020	Set the color space standard. <Remark> VC-A71PN does not support this feature
	Factory Reset	On / <u>Off</u>	Resume the factory default setting
Status			Display the current setting status

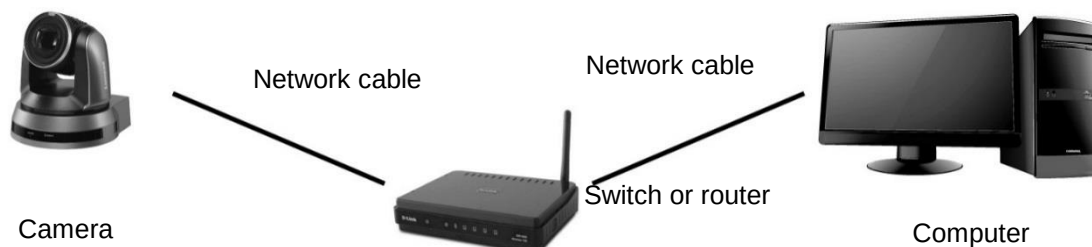
Chapter 6 Network Function Settings Description

6.1 Connecting Camera to Network

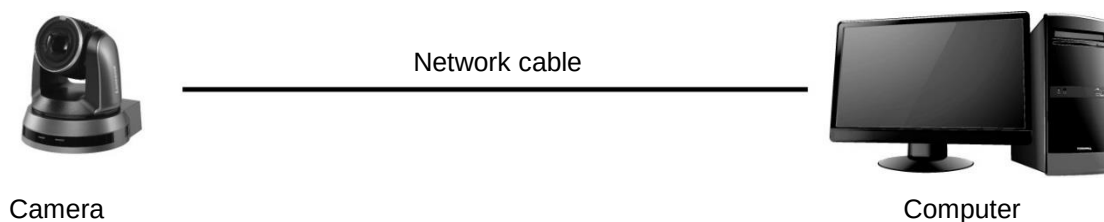
6.1.1 Connecting to the Internet

Two common connection methods are shown below

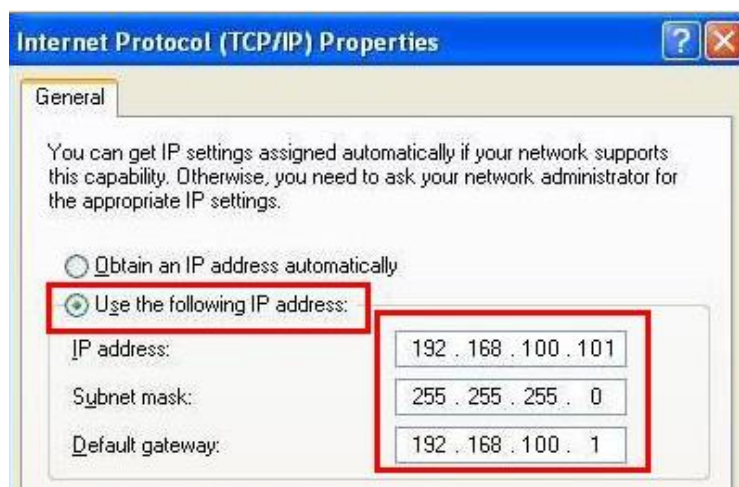
1. Connect the camera and the computer to the same switch or router



2. To connect the camera directly to a computer using a network cable, set the computer's IP address to the same network segment as the camera.
For example, the factory default IP address of the camera is 192.168.100.100. Set the computer's IP address to the same network segment, such as 192.168.100.101, to establish the connection.

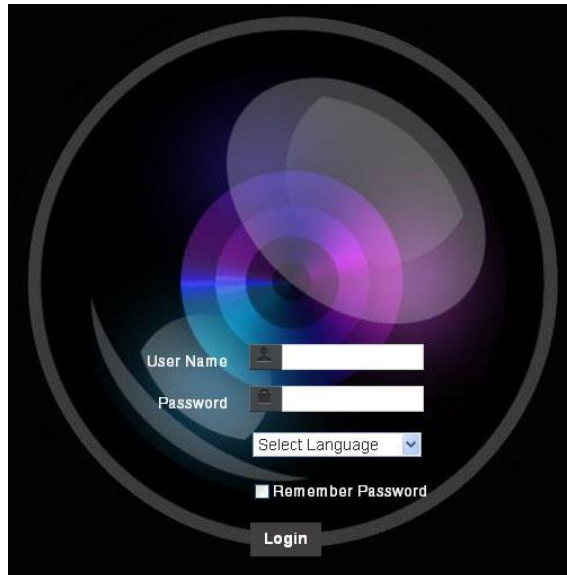


- Change network settings



6.1.2 Using the Browser to View the Images

- Open a browser, and enter the URL of camera in the IP address bar
e.g.: <http://192.168.100.100> (default IP address)
 - Enter administrator's account and password
- *For the first login, please refer to [6.2.3 Account Management](#) to change the default password



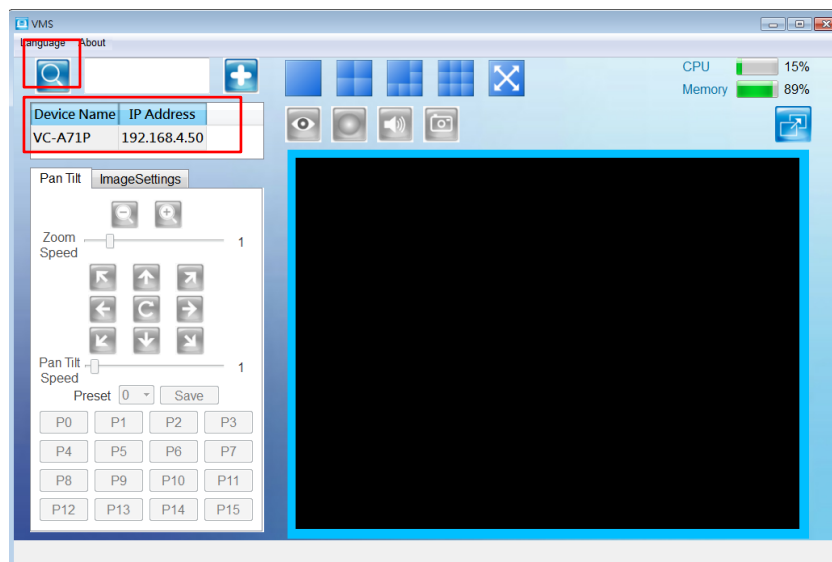
Lumens Network Camera	
●	Account: admin
●	Password: 9999

6.1.3 Using Lumens VMS Software to View the Images

<Remark> For VC-A71PN to support this function, please turn off NDI|HX (Please refer to [6.2.5 Setting-Video](#))

- Open **LUMENS VMS** software (Please download from the Lumens official website)
- Search for the camera: Press  [automatically search for the device] button to locate the VC-A71P camera
- Click the camera in the list and start operation after connecting to network

<Caution> When using automatic search, the camera and computer must be in the same network segment, e.g.: 192.168.4.X



6.1.4 Using RTSP Player to View the Images

<Remark> For VC-A71PN to support this function, please turn off NDI|HX (Please refer to [6.2.5 Setting-Video](#))

In addition to the browser and VMS, other free software also can be used for RTSP connection, such as VLC, Quick Time and PotPlayer

RTSP connection address formats are as follows:

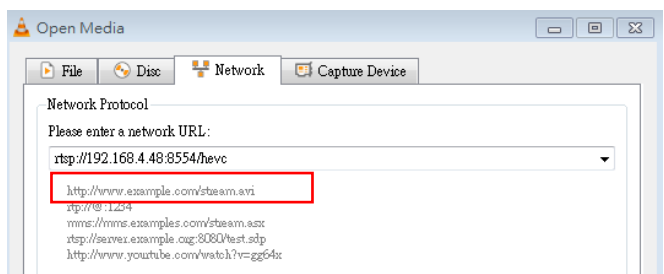
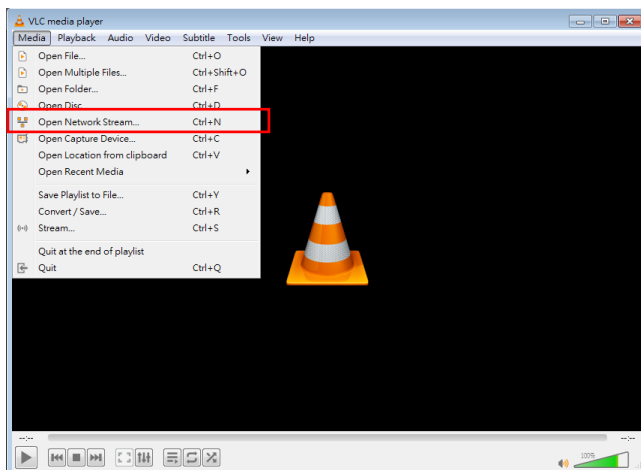
- RTSP Streaming 1 (4K@H.265) => rtsp://[camera IP:8554/hevc](#)
- RTSP Streaming 2 (1080P@H.264) => rtsp://[camera IP:8557/h264](#)
- RTSP Streaming 3 (720P@H.264) => rtsp://[camera IP:8556/h264](#)

For password authentication, the RTSP connection address is as follows:

- RTSP Streaming 1
[rtsp://Username:Password@VC IP address:port/hevc](#)
- RTSP Streaming 2/ 3
[rtsp://Username:Password@VC IP address:port/h264](#)
- To enable password authentication, please refer to [6.2.9 Setting - Network](#)

Example: Open VLC software, click [**Open Network Stream**], and enter a URL:

[rtsp://192.168.4.48:8554/hevc](#)



6.1.5 Using NDI Studio Monitor to View the Images

<Remark> Only VC-A71PN is available. Support when NDI|HX is on (Please refer to [6.2.5 Setting - Video](#))

For NewTek product operation or instructions, please visit the website:

<https://www.newtek.com/>.

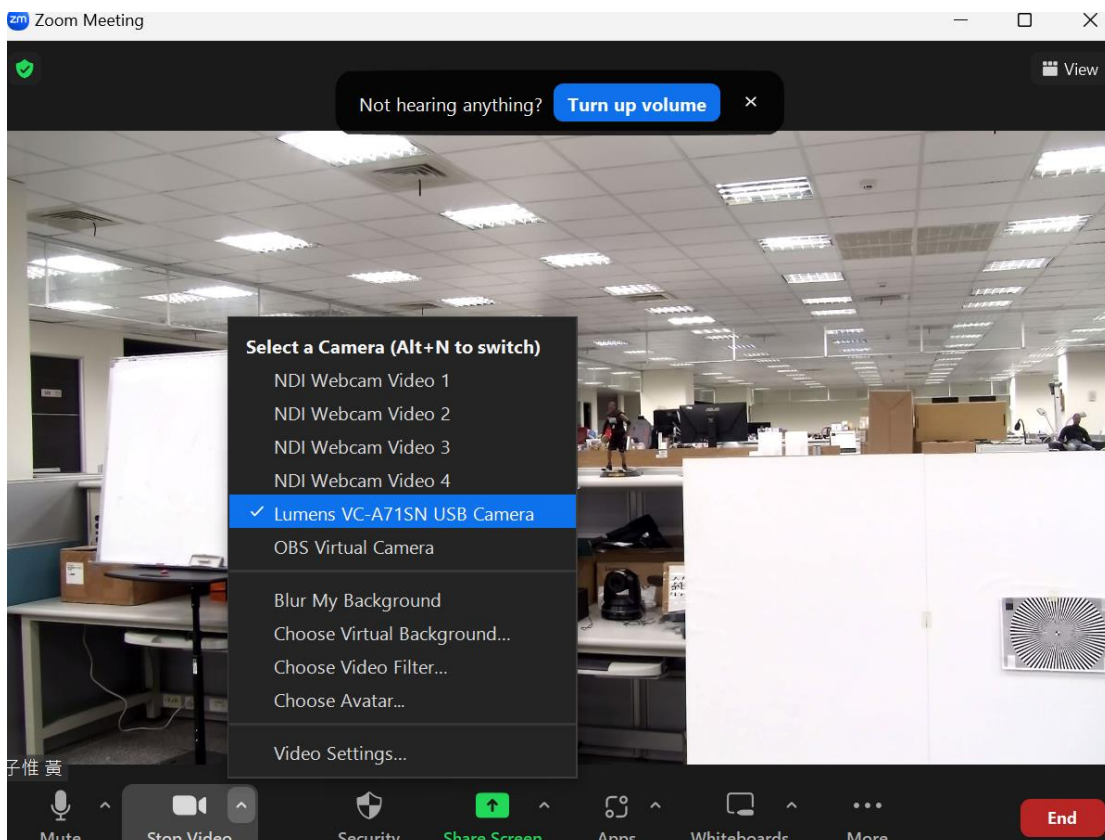
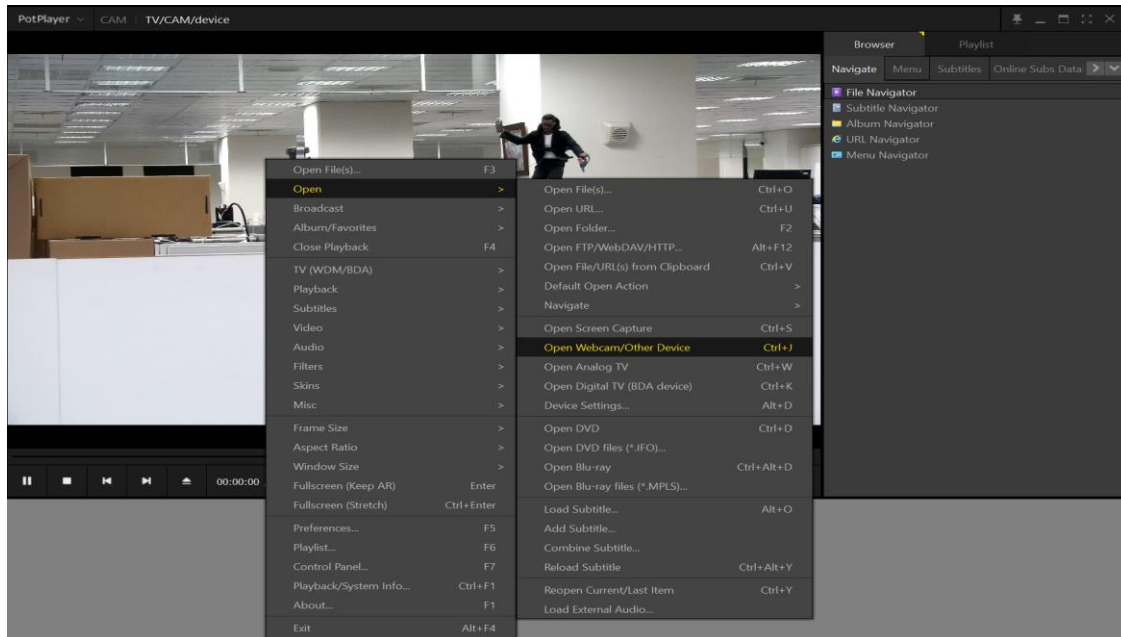
To download NDI 4 Tool, click the link: <http://new.tk/NDIHX-Driver-Win>

6.1.6 Using USB to View the Images

<Remark> When using 4K resolution for USB output, the video is encoded in H.264 format, while resolutions of 1080P and below are encoded in either H.264 or MJPEG format (the media player will choose automatically). However, some media players or

conference meeting apps may not display images when using 4K resolution due to conflicts between the A71P H.264 format and the software itself. In that case, please lower the resolution to 1080P.

To utilize the USB camera source, launch a media player like PotPlayer or a conference meeting app such as Zoom, then select the camera source.



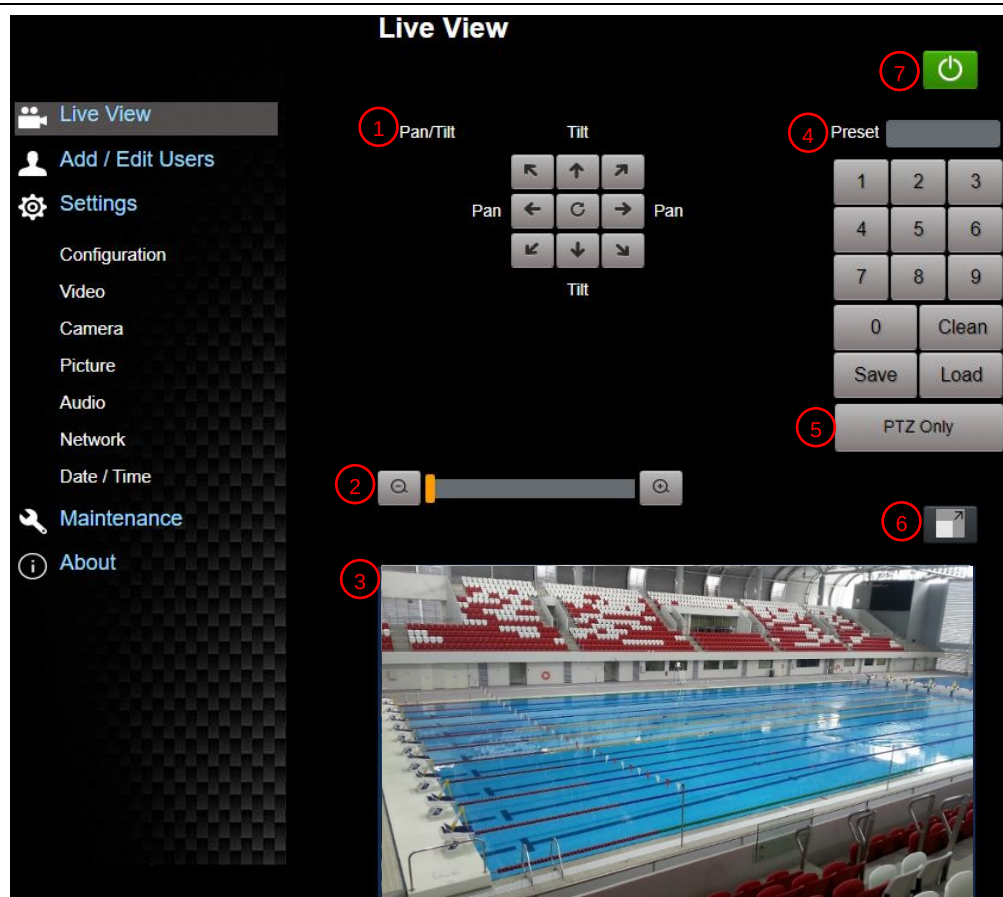
6.2 Web Page Function Description

6.2.1 Login Screen



No	Item	Function Descriptions
1	Username	Enter user account (default: admin)
2	Password	Enter user password (default: 9999) *For the first login, please refer to 6.2.3 Account Management the default password
3	Language selection	Currently, the system supports English, Traditional Chinese and Simplified Chinese
4	Remember password	Save user account name and password to the browser. When you log in next time, there is no need to re-enter them
5	Login	Log into the administrator screen on the website

6.2.2 Live View



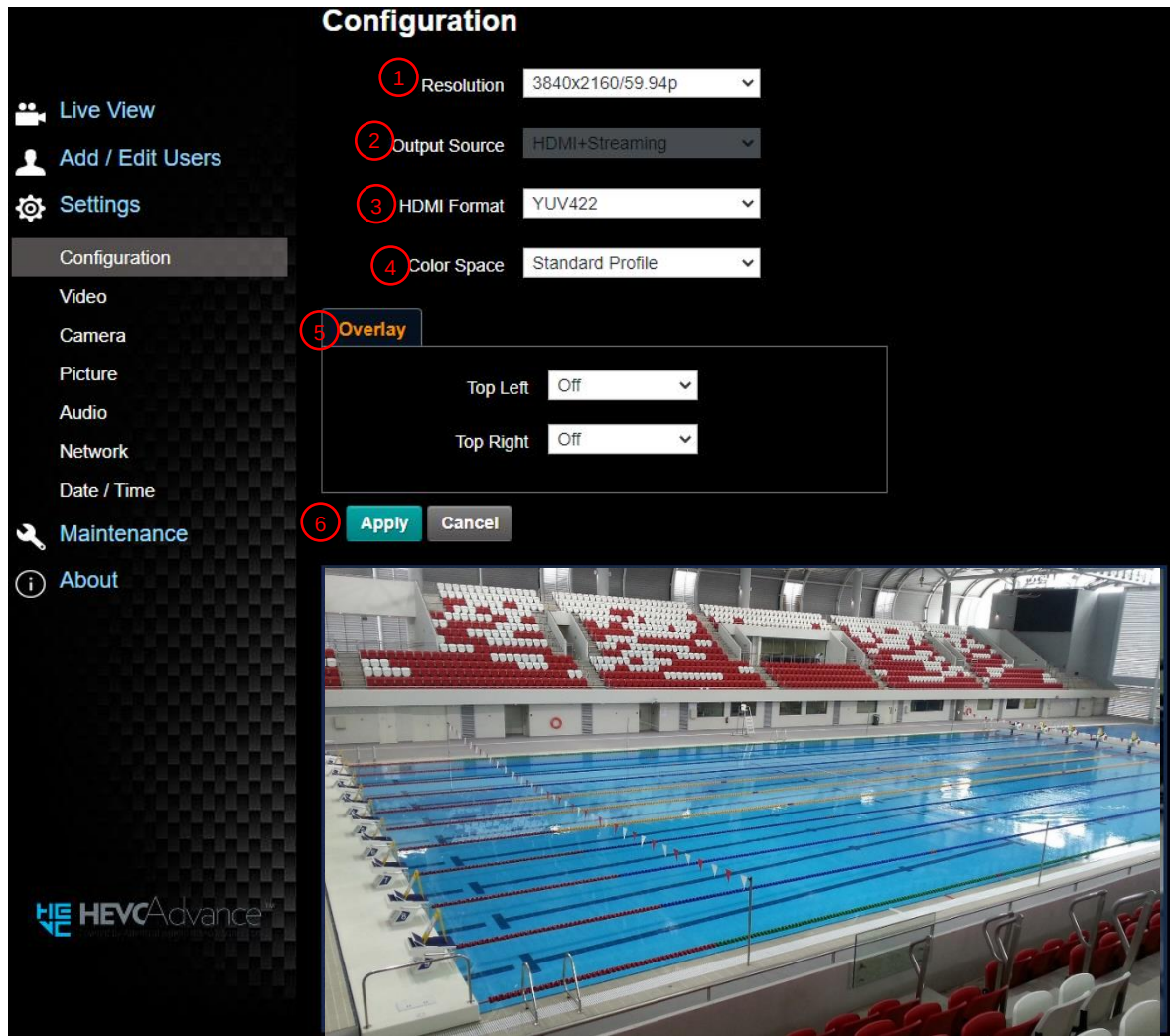
No	Item	Function Descriptions	VC-A71P	VC-A71PN
1	Pan / Tilt setting	Adjust the Pan/Tilt position of the camera screen	V	V
2	Zoom ratio	Adjust the zoom-in or zoom-out ratio via scroll bar	V	V
3	Preview window	Display the screen currently captured by the camera	V	Support when NDI HX is Off
4	Preset setting	Select the number first and then click SAVE or LOAD	V	V
5	PTZ Only	Enable/disable PTZ Only ▪ Disable (Default): Call the preset, execute Pan/Tilt/Zoom/Focus + picture parameters setting ▪ Enable: Call the preset, only execute Pan/ Tilt/ Zoom/ Focus	V	X
6	Switch to Full Screen	Switch the preview window to full screen	V	Support when NDI HX is Off
7	Power button	Turn the camera on or put it in standby mode.	V	V

6.2.3 Account Management

No	Item	Function Descriptions																
1	Add user account	Enter a user name and password to add a new user																
2	Permission setting	Set the new account management permissions <table border="1"> <thead> <tr> <th>User Type</th> <th>Admin</th> <th>Operator</th> <th>Viewer</th> </tr> </thead> <tbody> <tr> <td>View images</td> <td>V</td> <td>V</td> <td>V</td> </tr> <tr> <td>Settings</td> <td>V</td> <td>V</td> <td>X</td> </tr> <tr> <td>Account management</td> <td>V</td> <td>X</td> <td>X</td> </tr> </tbody> </table>	User Type	Admin	Operator	Viewer	View images	V	V	V	Settings	V	V	X	Account management	V	X	X
User Type	Admin	Operator	Viewer															
View images	V	V	V															
Settings	V	V	X															
Account management	V	X	X															
3	Applying setting	Add the newly created user to the list of account																
4	List of accounts	Edit: Modify the user password and permissions Delete: Delete the user account																

6.2.4 Setting - System Setting

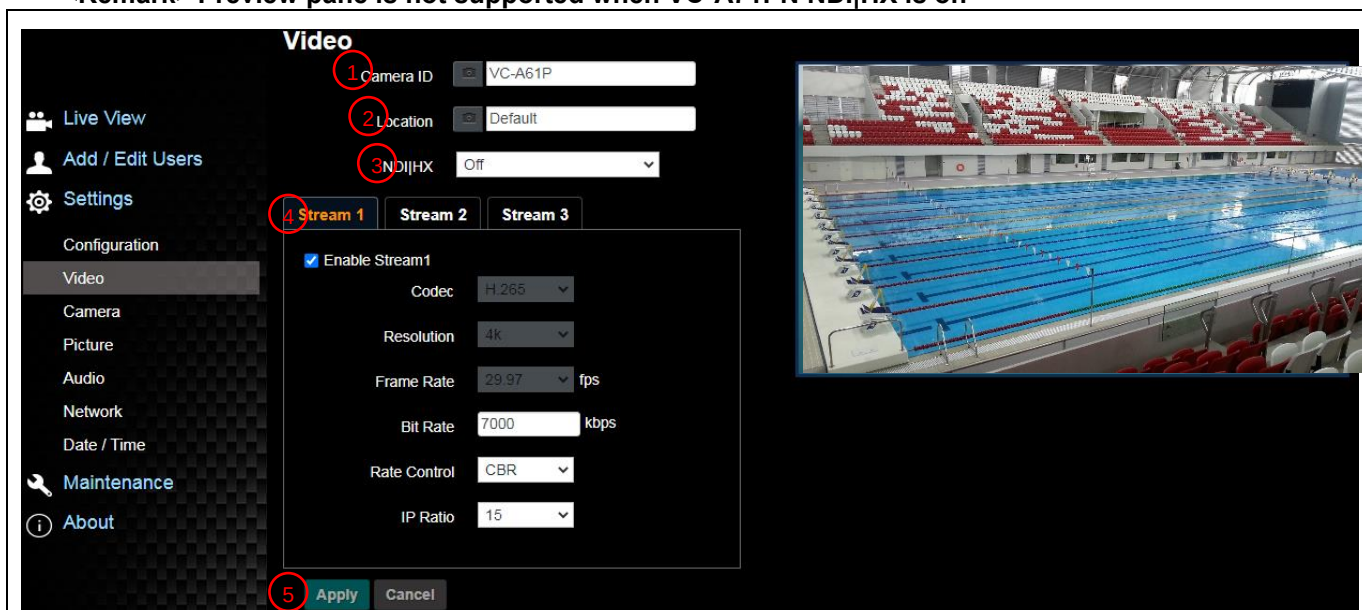
<Remark> Preview pane is not supported when VC-A71PN NDI|HX is on

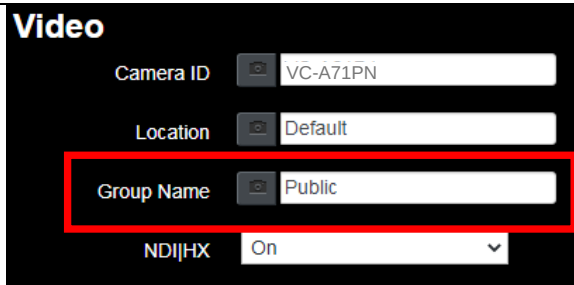


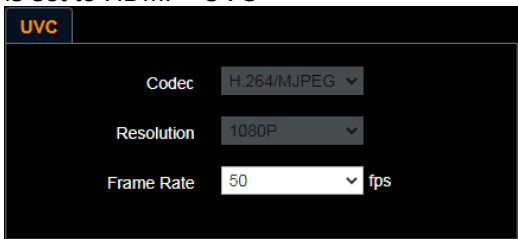
No	Item	Function Descriptions
1	Resolution	Set the resolution of the camera, resolutions supported by the camera are as follows: 3840 x 2160 59.94 / 50 / 29.97 / 25 fps 1080P 59.94 / 50 / 29.97 / 25 720P 59.94 / 50 / 29.97 / 25 1080i 59.94/50 After switching to the resolution, the camera will restart. Please refresh the browser <Remark>VC-A71PN does not support 720p 29.97/25 fps resolutions <Remark>1080i mode only supports HDMI/SDI output. When 1080i mode is set, NDI output is 1080p
2	Output signal	Select HDMI + streaming output or HDMI + UVC output <Remark> When it is set to HDMI+UVC output, 3840 x 2160 resolution only supports up 29.97p.
3	HDMI Format	Select YUV422 / YUV420 / RGB <Remark> When it is set to YUV420, 3840 x 2160 59.94p and 3840 x 2160 50p resolutions are supported
4	Color Space	Set the standard for color space Standard Profile/ REC.709/ BT.2020 <Remark> VC-A71PN does not support this feature
5	Overlay	Set the stream to display “date and time” or “custom content” and to display location
6	Apply	After the setting has been modified, select this button to apply the setting

6.2.5 Setting – Video

<Remark> Preview pane is not supported when VC-A71PN NDI|HX is on



No	Item	Function Descriptions	VC-A71P	VC-A71PN
1	Camera name	Modify the camera name - Camera names are limited to 1 - 12 characters - Please use a camera name by mixing uppercase and lowercase letters or numbers. Do not use “/” and “space” or special symbols <Remark> Modifying this field will modify the Onvif device name synchronously	V	V
2	Camera Location	Modify the location of the camera, such as Meeting Room 1 - Camera location is limited to 1 - 12 characters - Please use a camera name by mixing uppercase and lowercase letters or numbers. Do not use “/” and “space” or special symbols <Remark> Modifying this field will modify the Onvif device name synchronously	V	V
3	NDI HX	VC-A71PN can turn NDI HX on/off - On: Support NDI (ONVIF cannot find devices, and RTSP/RTMP/RTMPS/MPEG-TS/SRT streams cannot be used.) - Off: Support RTSP/RTMP/RTMPS/MPEG-TS/SRT, and ONVIF can find devices but cannot be used with NDI Tool		V
3.1	Group Name	Video  The group name can be modified here and set with Access Manager - Receive in NDI Tool If no group is assigned, the machine belongs to Public. As		Support when NDI HX is On

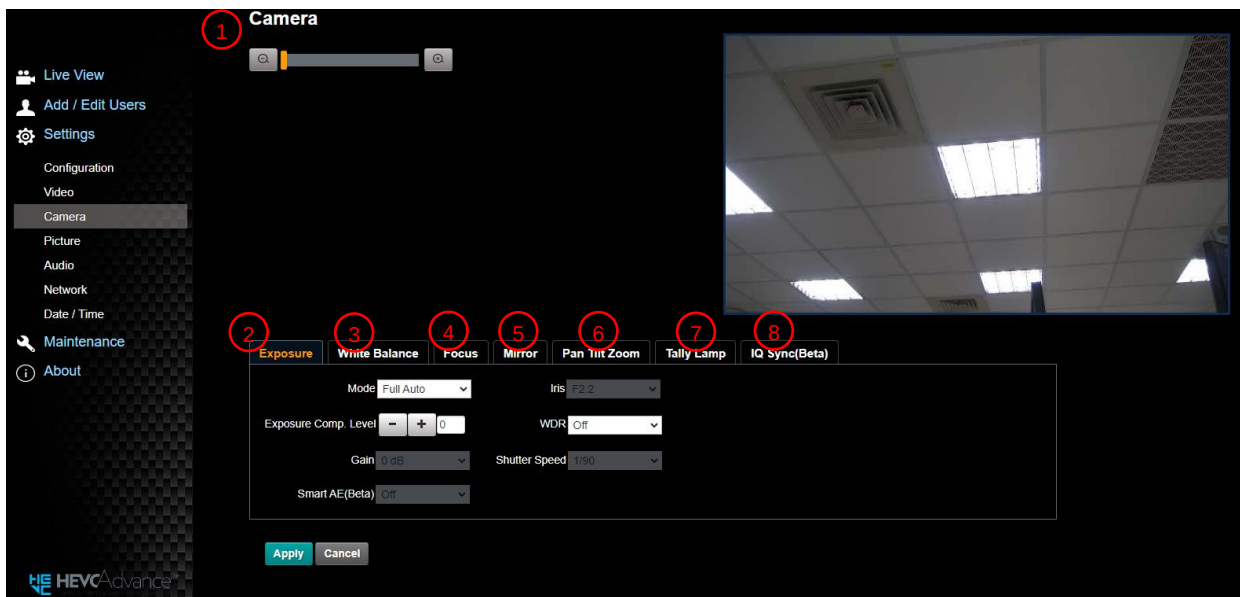
		the default of Access Manager includes Public, Studio Monitor can discover this machine. ■ If a group is assigned, but this group is not added into Access Manager, Studio Monitor cannot discover this machine. ■ If a group is assigned and this group is added into Access Manager, Studio Monitor can discover this machine. The name can contain commas (,) to distinguish different groups Example: "default, 123, abc" means this machine belongs to three groups at the same time (default / 123 / abc)		
4	Streaming 1/ Streaming 2/ Streaming 3	<u>Setting - System Setting - Output Signal</u> Display when it is set to HDMI + streaming VC-A71P supports 3 streaming outputs. Please refer to <u>6.2.5.1 Streaming Parameter Setting</u> for relevant settings	V	Support when NDI HX is On
4.1	UVC	<u>Setting - System Setting - Output Signal</u> Display when it is set to HDMI + UVC  ■ Encode Format: ➢ 3840 x 2160 29.97 / 25 fps: H.264 ➢ Less than 1080p (inclusive): H.264/MJPEG ■ Resolution: It is shown based on <u>Settings - System Settings - Resolution</u> Frame Rate: 3840 x 2160 resolution only supports up to 29.97 fps	V	
5	Apply	After the setting has been modified, select this button to apply the setting	V	V

6.2.5.1 Streaming parameter setting

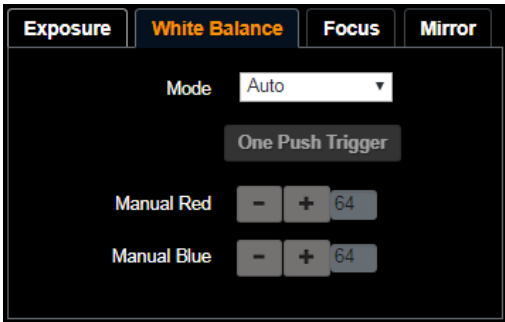
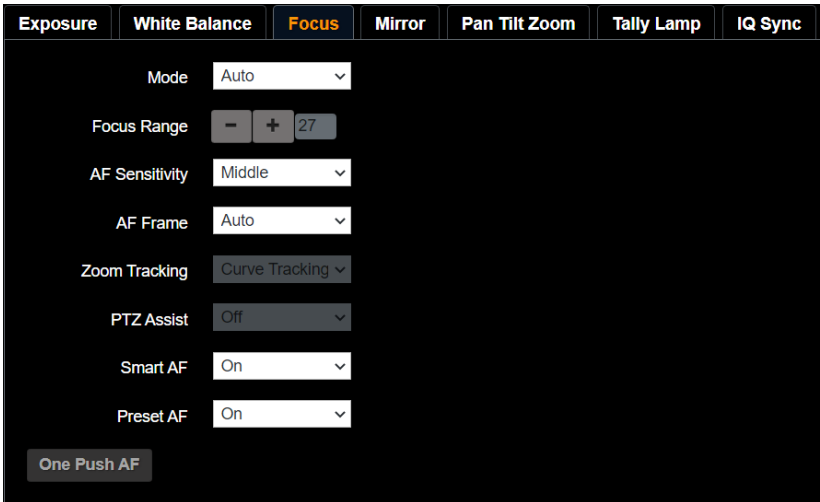
Function		Streaming 1	Streaming 2	Streaming 3
Encode Format		H.265	H.264	
Resolution		4K /1080p / 720p	1080p / 720p / 1080i	640 x 360
Frame Rate		Setting according to the supported resolution <Remark> Only 59.94 / 50 are supported when the resolution of Streaming 2 is 1080i		
Bit Rate (kbps)	2,000 ~ 20,000	2,000 ~ 20,000	512 ~ 5,000	512 ~ 5,000
	7,000	15,000	7,500	1,000
Rate Control		CBR / VBR		
IP Ratio		Setting according to the supported resolution <Remark> Only 15 / 25 are supported when the resolution is 1080i		

6.2.6 Setting – Camera

<Remark> Preview pane is not supported when VC-A71PN NDI|HX is on



No	Item	Function Descriptions
1	Zoom ratio	Adjust the zoom-in or zoom-out ratio via scroll bar
2	Exposure	■ Mode: Select exposure mode (Automatic/Shutter Priority/Aperture Priority/Manual) ■ Exposure Comp. Level: Select exposure compensation level ■ Gain: The gain limit is adjustable when the exposure mode is set to "Manual" ■ Iris: The size of aperture is adjustable when the exposure mode is set to "Manual" or "Aperture Priority" ■ WDR: Set the level of wide dynamic range (WDR) in order to obtain better images ■ Shutter Speed: The shutter speed is adjustable when the exposure mode is set to "Manual" or "Shutter Priority" ■ Smart Exposure: Enable/Disable Smart Exposure; Available when "Focus - SMART AF" is on (Currently it's the beta version. Please refer to Web Page Firmware Update Information for the official version release schedule.)

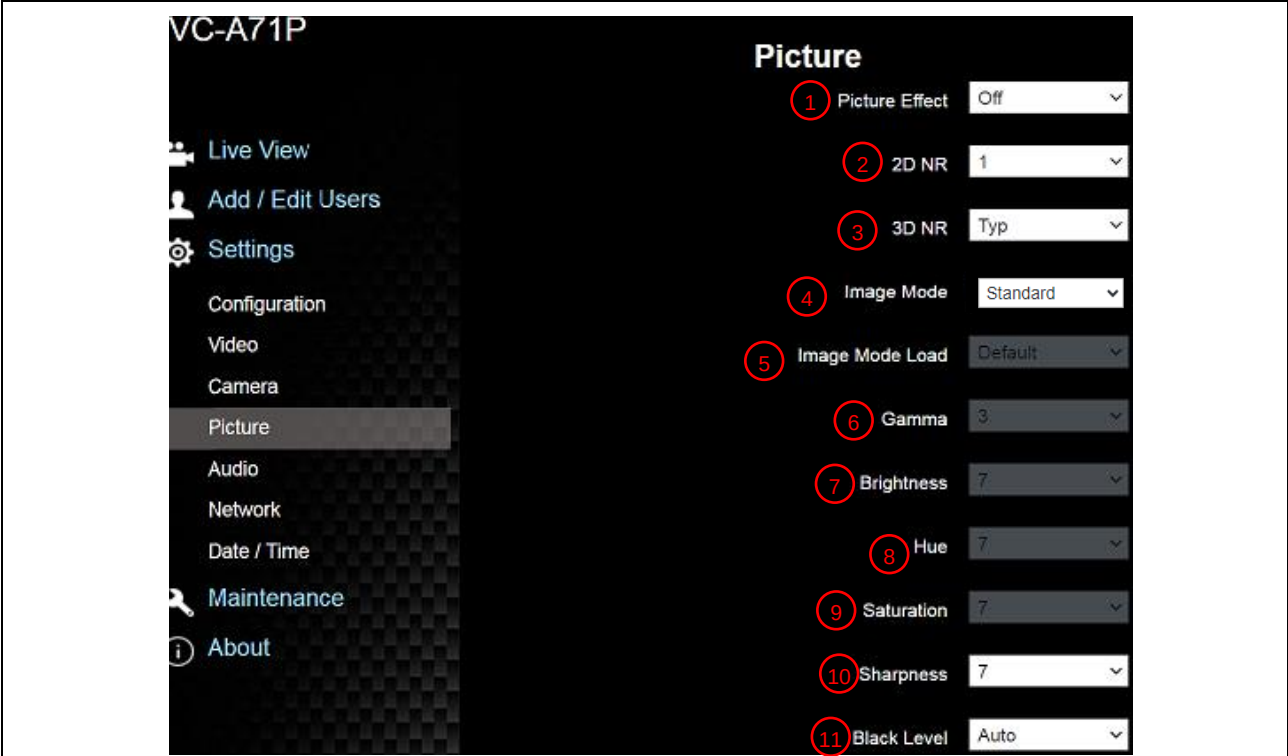
3	White Balance	 ■ Mode: Select the color temperature mode ■ Open Push WB: One push color temperature is adjustable when the white balance mode is set to “One Push WB” ■ Manual Red/Blue: Manually adjust blue/red color temperature
4	Focus	 ■ Mode: Select manual/automatic focus ■ Focus Range: The focusing range is adjustable when the focus mode is set to “Manual” ■ AF Sensitivity: Set automatic focus sensitivity ■ AF Frame: Set automatic focus range ■ Zoom Tracking: Set the focus mode during lens zooming in Manual Focus mode ■ PTZ Assist: Set Auto Focus in manual mode ■ Smart AF: Turn on this function and AF will mainly focus on the face ■ One-shot Focus: When manual focus mode is on, click this function for one-shot autofocus
5	MIRROR	 ■ Mirror: Set automatic flip mode [Off / Mirror / Flip / Mirror + Flip]

6.	Pan Tilt Zoom	■ Pan/Tilt Limit: Set the Pan/Tilt limit ■ Pan/Tilt Flip: Set Pan/Tilt flip ■ Preset Speed: Set the rotation speed of the cradle head when Preset is executed ■ PTZ Speed Comp: Set the Pan/Tilt moving speed to vary from the zoom position ■ Initial Position: Set the camera lens to return to the last operated position or the first preset position after POWER ON <Remark>First default position = Preset 0 ■ D-Zoom Limit: Set the D-Zoom Limit, x1 - x12 ■ Motionless Preset: When the function is enabled, the screen will Freeze when Preset is executed. Freeze will be released after Preset is completed ■ PTZ Motion Sync: After turning on, operate Preset to achieve PTZ Motion Sync <Remark> This function will refer to the default speed setting, but may automatically adjust the speed due to the difference in PTZ movement range
7	Tally Lamp	■ Enable/Disable Tally Lamp ■ Indicator Light Link Mode: Set whether the indicator light is linked with RS232 command ➢ General (Not linked): The indicator light switch and mode shall be set separately. -When the indicator light is on and the default mode is off, additional command is required to set the indicator light mode -Setting indicator light mode when the indicator light is off will not trigger

		the indicator light ➤ Link: -When the indicator light is on, the default mode is red light -Setting indicator light mode when the indicator light is off will turn on the indicator light
8	IQ Sync	 Set IQ main synchronized device Available when “Exposure” and “White Balance” is set to “Auto” mode (Currently it's the beta version. Please refer to Web Page Firmware Update Information for the official version release schedule.)

6.2.7 Setting – Picture

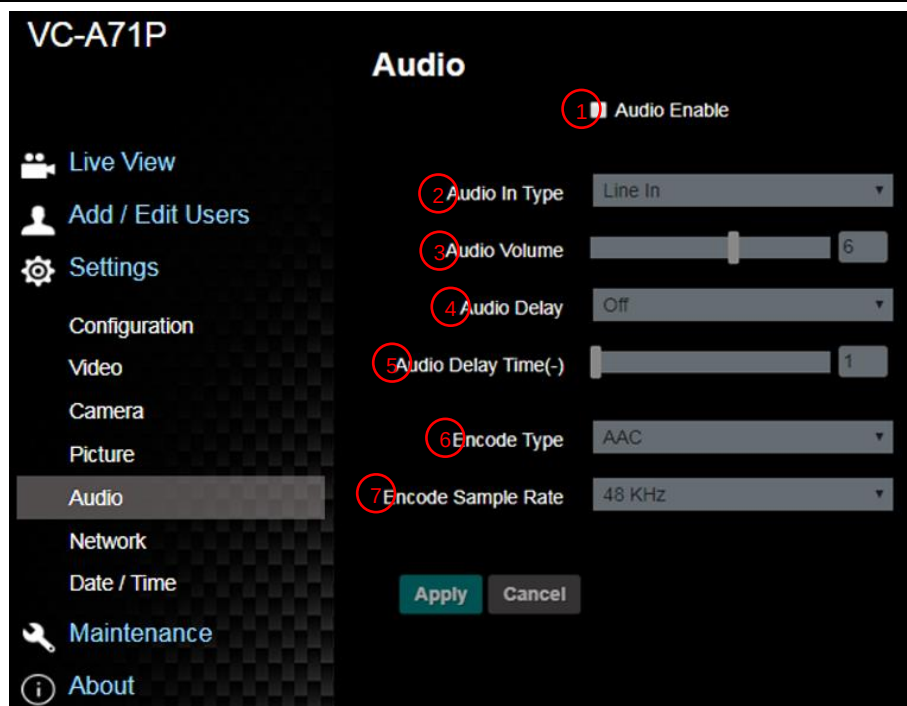
<Remark> Preview pane is not supported when VC-A71PN NDI|HX is on



No	Item	Function Descriptions
1	Picture Effect	Set picture effect, Off/Neg/Black and White
2	2D noise reduction settings	2D noise reduction settings
3	3D noise reduction settings	3D noise reduction settings
4	Image mode	The user may customize his/her desired image mode.
5	Image Mode Load	When no custom setting is needed, reset the picture parameters back to the factory default by selecting this item
6	Gamma	Gamma Level adjustment; Adjustable when the image mode is set to Custom
7	Brightness	Brightness adjustment; Adjustable when the image mode is set to Custom

8	Contrast	Contrast adjustment; Adjustable when the image mode is set to Custom
9	Saturation	Saturation adjustment of the image; Adjustable when the image mode is set to Custom
10	Sharpness	Adjust the sharpness of the image
11	Black Level	Adjust the brightness level of dark areas

6.2.8 Setting - Audio



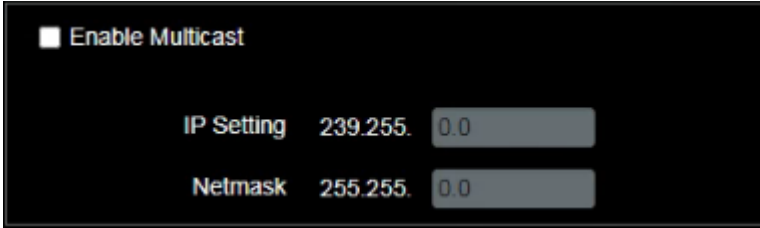
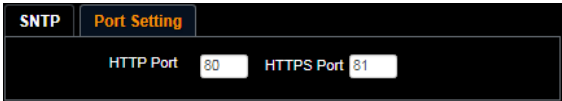
No	Item	Function Descriptions	VC-A71P	VC-A71PN
1	Open audio	Turn on/ off sound	V	V
2	Soundtrack effect setting	Set MIC In/Line In	V	V
3	Audio Volume	Adjust Volume	V	V
4	Audio Delay	Turn on/ off Audio Delay	V	Support when NDI HX is Off
5	Audio Delay Time(ms)	Set Audio Delay Time (-1~-500ms)	V	
6	Encode Type	AAC/ G.711	V	
7	Encode sample rate	Set Encode sample rate ■ 48 KHz (AAC) ■ 44.1 KHz (AAC) ■ 16 KHz (AAC) ■ 16 KHz (G.711) ■ 8 KHz (G.711)	V	

6.2.9 Setting - Network

The screenshot shows the 'Network' settings page. On the left is a sidebar with navigation options: Live View, Add / Edit Users, Settings (selected), Configuration, Video, Camera, Picture, Audio, Network, Date / Time, Maintenance, and About. The main content area is titled 'Network' and contains several sections:

- 1. Network:** Includes DHCP (On), IPv4 Address (192.168.4.37), IPv4 Netmask (255.255.255.0), IPv6 Address (2001:2001:2001:2001), IPv6 Netmask (2001:2001:2001:2001), Gateway (192.168.4.254), DNS (192.168.1.211), and MAC Address (dc:62:ac:01:04:9e). An 'Apply' button is at the bottom.
- 2. RTMP / RTMPS:** Includes 'Publish to Server' (checked), Video encoder type (H.264@7000Kbps), Audio encoder type (None), Resolution format (1080P(29.97)), Chunk size (1412), RTMP Server Status (RTMP disable), RTMP URL, Stream Key, and 'Require Password Authentication' (checked) with fields for Account and Password.
- 3. RTSP:** Includes 'Enable Multicast' (checked) and 'Require Password Authentication' (checked).
- 4. MPEG-TS:** Includes 'Enable MPEG-TS' (checked), Stream Source (Stream 2), and MPEG-TS IP (192.168.4.37) with Port 0.
- 5. SRT:** Includes 'Enable SRT Streaming' (checked), Stream Source (Stream 2), Connection Mode (Caller), URL, Stream ID, Latency (120 ms), Encryption (None), Passphrase, and Connect Status (SRT Disable).
- 6. SNTP:** Includes SNTP Server (pool.ntp.org).
- 7. Port Setting:** Includes Port (0).
- 8. Tracking Data Out:** Includes 'Enable' (checked), Serial Output (selected), IP Output (deselected), and Destination IP (0.0.0.0) with Port 0.

No	Item	Function Descriptions	VC-A71P	VC-A71PN
1	Network	Network setting of camera. Change of setting is available when DHCP function is closed.	V	V
2	RTMP Setting	Copy the RTMP web address provided by the RTMP service platform and paste it to the RTMP connection address to publish the camera images on the RTMP service platform To upload to YouTube for live streaming, the audio function must be turned on first To upload to YouTube for live streaming, please see 6.2.8 Setting - Audio to turn on the audio function first	V	
3	RTSP Setting	■ Enable / Disable Multicast When more than 4 people are connected to watch live video at the same time, it is recommended to enable the multicast function ■ Enable / Disable Account Password Authentication ➤ Refer to 6.1.4 Using RTSP Player to View Images for RTSP connection format ➤ The account password is the same as that for the camera web login page. Please refer to 6.2.3 Account Management to add/revise the account information	V	Support when NDI HX is Off

3-1	Multicast Setting	The setting can be configured after Multicast is enabled 		Support when NDI HX is On
4	MPEG-TS Setting	Set MPEG-TS format <Remark> The port setting range 1025 ~ 65535 <Remark> The following port has been used by the camera. Setting of the port may not connect correctly 1935, 8080, 8554, 8556, 8557, 9090	V	Support when NDI HX is Off
5	SRT Setting	- Set the field of SRT and then check the item to open SRT streaming After the SRT streaming is opened, it will be connected automatically upon startup - The port setting range 1025 ~ 65535 <Remark> The following port has been used by the camera. Setting of the port may not connect correctly 1935, 8554, 8555, 8556, 8557 - Delay time is for 20 to 8000 microseconds. The default value is 120 microseconds	V	Support when NDI HX is Off
6	SNTP Setting	Set the SNTP server address	V	V
7	Port Setting	 Set HTTP port. The default Port value is 80 Set HTTPS port. The default Port value is 81 <Remark> The settings can only be modified when the camera's DHCP function is turned off	V	V
8	Tracking Data Output Setting	When enabled, PTZ position information can be fed back when the camera is moving - Serial port output: The protocol must be set to VISCA, and the baud rate should be 38400 (For related settings, please refer to 6.2.14 Setting - Maintenance - System Service) - IP output: It can set IP address and communication port - The port numbers should be > 1024. 1935, 8555, 8556, 8557 cannot be used. <Remark> The performance of the camera may be affected when Tracking Data Output is enabled	V	V

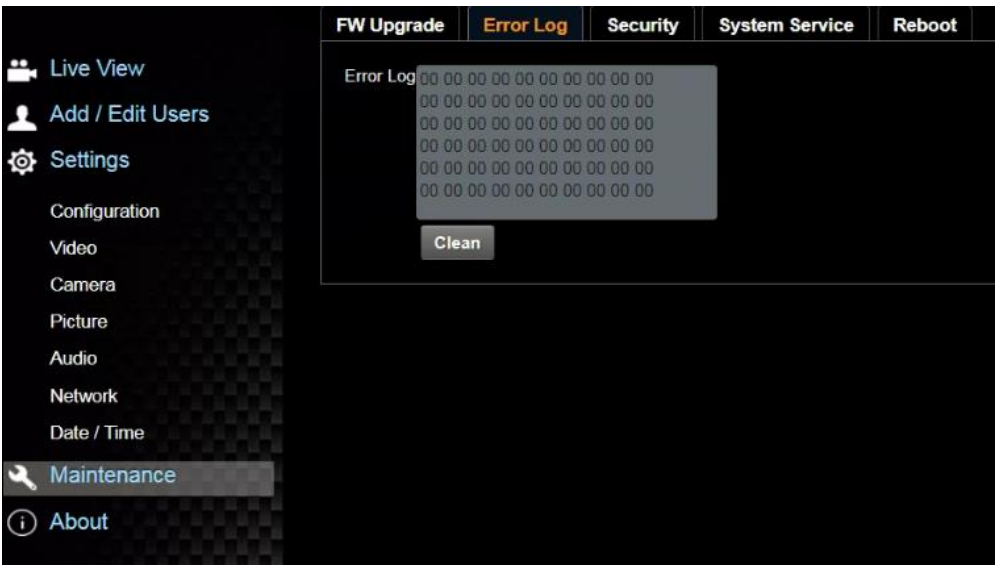
6.2.10 Setting - Time

No	Item	Function Descriptions
1	Camera Time	Display the date and time of the camera
2	Set the Time	- Manual Configuration Set time manually - Synchronize with computer time: Set the camera time according to the computer time - Synchronize with SNTP server: Set the camera time synchronously with the SNTP server <Remark>SNTP server address: Please change in network setting

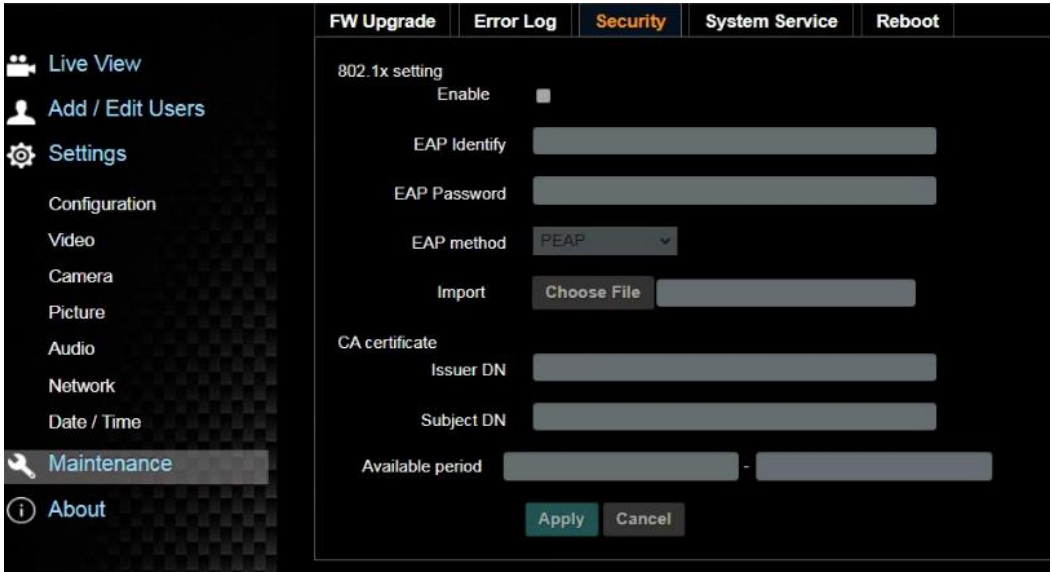
6.2.11 Setting - Maintenance - Upgrading Firmware

No	Item	Function Descriptions
1	Firmware Upgrade	The camera firmware may be upgraded via web page. For the upgrade method, please download the FW upgrade manual from Lumens official website <Remark> Please do not operate or turn off the power of the device during the update to avoid firmware update failure

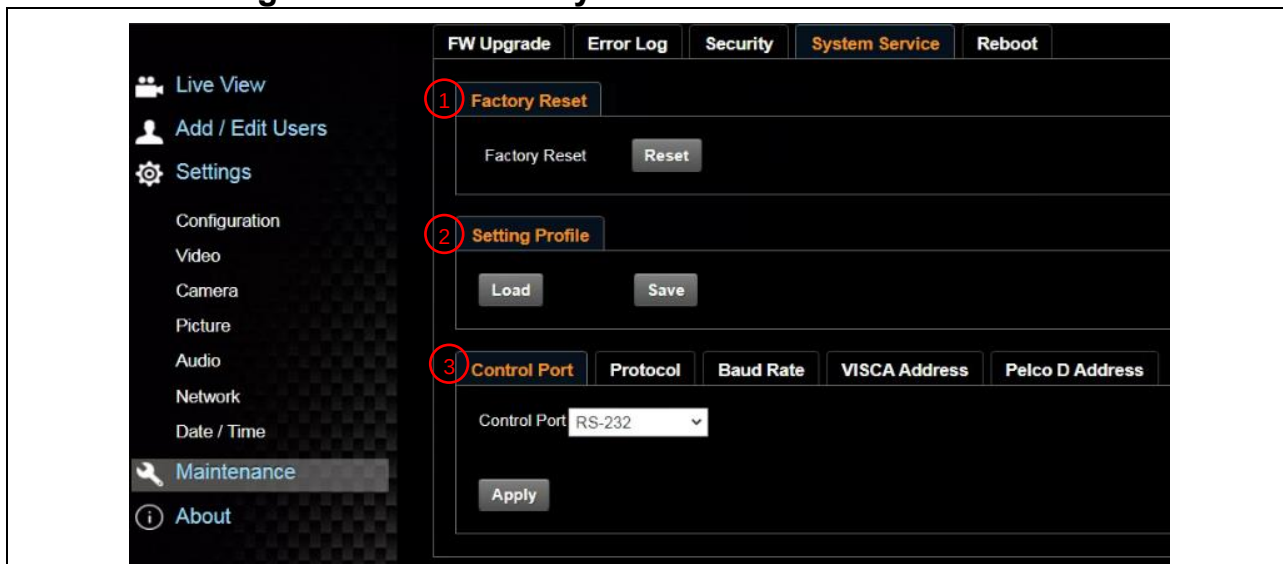
6.2.12 Setting - Maintenance - Incident Log

		
No	Item	Function Descriptions
1	Event Log	If the camera encounters errors, an error code log will be established <Remark> When an error code appears, please try to clear it to make sure whether the issue has occurred repetitively

6.2.13 Setting – Maintenance - Security

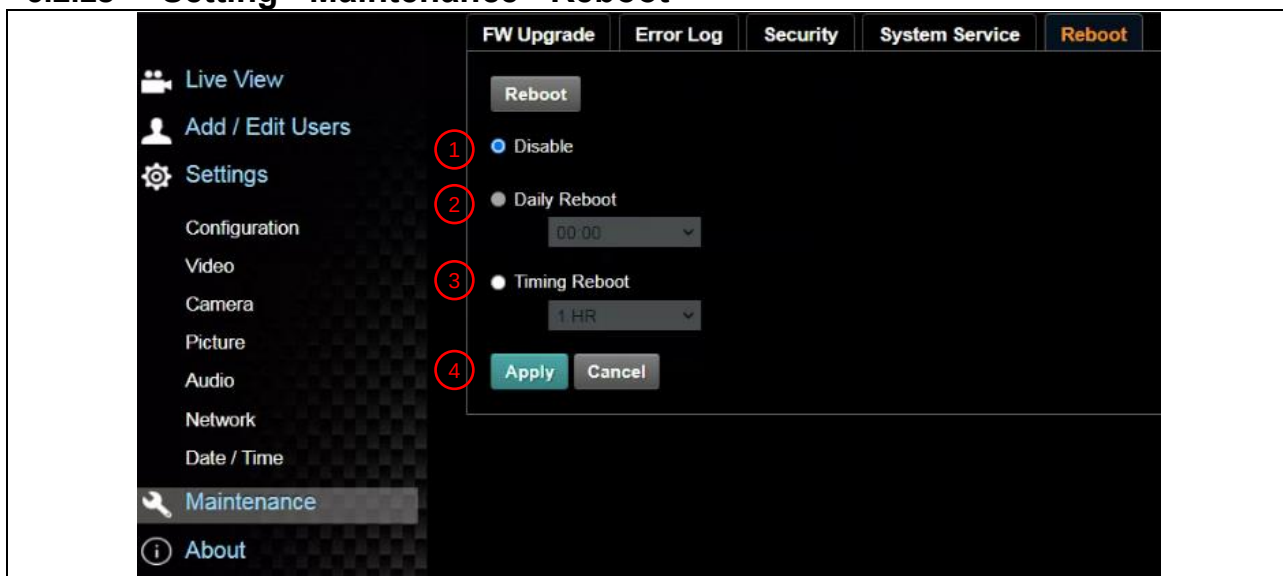
		
No	Item	Function Descriptions
1	802.1x Setting	Enable/Disable 802.1x Protocol. Setting can be made after enabling. <Remark> In order to enable this function, the router must support 802.1x Protocol. <Remark> VC-A71PN does not support this function.

6.2.14 Setting - Maintenance - System Service



No	Item	Function Descriptions
1	Factory Reset	Select the reset button to resume the factory default setting as the web page setting
2	Setting Profile	The web page setting parameters can be exported from the computer and imported/applied to another camera
3	Control Port	Communication protocol related settings for connection with RS-232/RS-422 communication

6.2.15 Setting - Maintenance - Reboot



No	Item	Function Descriptions
1	Reboot	Reboot immediately
2	Disable	Disable the reboot settings
3	Daily reboot	Set the daily reboot time <Remark> Please go to Settings - Time to complete the SNTP server synchronization to set daily reboot
4	Timing Reboot	Set the timing reboot time

6.2.16 About

No	Item	Function Descriptions
1	Camera ID	Display the camera name
2	Firmware Version	Display the firmware version of the camera
3	Detail Information	Display firmware version-related information
4	Serial Number	Display the camera serial No.
5	Uptime	Displaying the running time of the camera (the timing of standby continues; the time will be reset when the camera is rebooted or turned off)

Chapter 7 DIP Switch Setting

7.1 DIP SWITCH

7.1.1 OUTPUT Switch (Switch to different output resolutions)

To switch via the OSD/ RS-232 Command is also available based on the last executed action



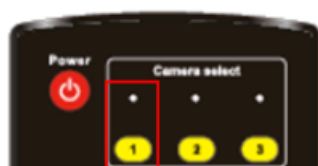
ID	Resolution	Frame Rate	
0	3840x2160p	59.94	
1		50	
2		29.97	
3		25	
4	1920x1080p	59.94	
5		50	
6		29.97	
7		25	
8	1280x720p	59.94	
9		50	
A		29.97	<Remark> VC-A71PN does not support 720p 29.97/25 fps resolutions
B		25	
C	1920x1080i	59.94	<Remark> 1080i mode only supports HDMI/SDI output. When 1080i mode is set, NDI output is 1080p
D		50	
E	Reserved	Reserved	
F			

7.1.2 IR SELECT (Selected by Remote control)

When using the remote control, please make sure the settings of IR SELECT and the Camera Select of the remote control are consistent



ID	Setting
1	
2	
3	



Chapter 8 Troubleshooting

This chapter describes problems you may encounter while using VC-A71P/VC-A71PN. If you have questions, please refer to related chapters and follow all the suggested solutions. If the problem still occurred, please contact your distributor or the service center.

No.	Problems	Solutions
1.	Boot without power signal	1. Make sure you have plugged in the power cord. 2. When using a PoE connection, ensure that the power supply supports PoE+ / PoE++ hubs ■ VC-A71P : PoE++ (IEEE802.3bt) ■ VC-A71PN : PoE+ (IEEE802.3at)
2.	There is no image output from VC-A71P	1. Check the power supply or PoE supply functions. 2. Confirm the output signals are in streaming output. 3. Confirm whether the camera resolution can be used together with the monitor equipment ■ If 4k output, confirm the monitor equipment supports 4K output 4. Replace the cables and make sure they are not faulty.
3.	RS-232 cannot be controlled	1. Confirm the connection is correct (RS-232 In/ Out) 2. Please make sure the Baud rate setting is the same as the control equipment
4.	Whether the Internet can be used for operation	Please refer to Chapter 6 Network Function Settings Description for the Internet usage
5.	ONVIF software cannot locate the machine	Please be sure to use only English alphabets or numbers in [Setting]>[Video]> [Camera](Location) of the webpage. Using special characters and pressing the spacebar will cause the ONVIF software unable to locate the machine. <Remark> For VC-A71PN, either NDI or ONVIF can be used. Please go to the [Setting] > [Video] > [NDI HX] on the web page to check if it is [Off].
6.	Tally lamp is not working when applying RS232 setting	Please make sure if the Tally lamp function is enabled from the setting menu or web page. ■ Setting Menu: System - Tally Lamp ■ Web page: Setting - Camera - Tally Lamp
7.	The camera does not save the relevant parameters (PTZ, AWB ...) after reboot	Please make sure if the Initial Position is set to Last Operated Position from the setting menu or web page ■ Setting Menu: System - Initial Position ■ Web page: Setting - Camera - Pan Tilt Zoom
8.	Tracking Data Output is delayed or not updated	Please make sure that the camera is not in the following conditions. The following function settings may cause data output to be delayed or not updated:

		- Modify output resolution - Modify camera mode - Modify HDMI format - Enable OSD menu - Streaming setting - Audio setting
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Supplier's Declaration of Conformity 47 CFR § 2.1077 Compliance Information

Manufacturer : Lumens Digital Optics Inc.

Product Name : VC-A71P/ VC-A71PN

Model Number : 4KPTZ Video Camera

Responsible Party – U.S. Contact Information

Supplier : Lumens Integration, Inc.

4116 Clipper Court, Fremont, CA 94538, United States

e-mail : support@mylumens.com

FCC Compliance Statement

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions : (1) This device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.